

HW01 Report — QA/QC Jobs · 20 Defects · Physical Product Testing

StudentID: 23127075
Student Name: Lê Trung Kiên
Course: Software Testing
Exercise: HW01
Submission Date: 04/06/2026

1) Requirement 1 — QA/QC Job Market 2026+ (40 pts)

1.1 Summary

- Total postings collected: 10/10
- Published within 60 days before submission: Yes
- Postings requiring AI/LLM/automation-AI skills: 9

1.2 Job Postings

1.2.1 Posting 01

- Job title: Business Analyst/Quality Assurance Analyst
- Company: Vion Consultants, Inc
- Location: Remote (primarily remote; occasional on-site in Baltimore, MD)
- Platform: SimplyHired
- Posting date: 01/06/2026
- Job URL:
<https://www.simplyhired.com/job/sZlt5OX6ynl8dQqiHuvfhuxrGj2L6sjsjSvNIQHlXln0TRQetvLm8g>
- Salary: From \$40.00 an hour
- Requires AI/LLM/automation-AI skills? Y (manual + automation QA context)

Job Description

Full Job Description

Location: Primarily Remote (some on-site visit may be required in Baltimore, MD)

Duration: 9 Months (possibility of extension)

Candidate Location: Candidates within the DC/MD/VA metro area are preferred

Job Summary

We are seeking a dynamic and detail-oriented business analyst/quality assurance analyst to join our innovative technology team. In this role, you will be at the forefront of ensuring the quality and functionality of our software solutions, collaborating closely with developers, business analysts, and

stakeholders. Your expertise will drive the development of robust test plans, execute comprehensive testing strategies, and facilitate seamless communication across teams to deliver high-quality products. If you thrive in a fast-paced environment and are passionate about software quality assurance and business analysis, this is your opportunity to make a meaningful impact.

Responsibilities

- Maintain requirements traceability across Strategic Brief, IA, and UX deliverables.
- Document IA requirements and validate user-flow definitions.
- Conduct stakeholder interviews and capture structured research outputs.
- Track content workbook entries and validate retain/edit/remove dispositions.
- Define mockup acceptance criteria and operate the defect log.
- Trace UX Recommendations back to research and data points.
- Serve as primary QA lead across testing, defect management, and pre-launch verification.
- Author test plans and test cases for content edits, migration, redirects, and site testing.
- Validate content edits across up to 500 high-priority pages.
- Perform migration validation on a per-page basis.
- Conduct WCAG 2.1 AA accessibility QA and remediation verification.
- QA the redirect mapping for all 500 pages.
- Manage hyper-care defect intake, prioritization, and triage.

Required Qualification

- Minimum 7 years of combined Business Analysis and QA experience.
- Experience with content audits and content management workflows.
- Strong manual testing experience with exposure to automated testing.
- Proficient with defect tracking tools (Jira, Azure DevOps, or equivalent).
- Demonstrated accessibility testing exposure.

Preferred Qualifications

- ISTQB or CSTE certification.
- Drupal-based testing experience.
- Experience validating migrated content at scale.
- Hands-on experience with WCAG/Section 508 tooling (axe, WAVE, NVDA, JAWS).

Pay: From \$40.00 per hour

Application Question(s)

- Will you now or in the future require sponsorship to work in the US?
- Do you have experience performing accessibility testing?
- Do you have experience with Jira, Azure DevOps, or equivalent?

License/Certification: ISTQB or CSTE certification (Preferred)

Work Location: Hybrid remote in Baltimore, MD 21202

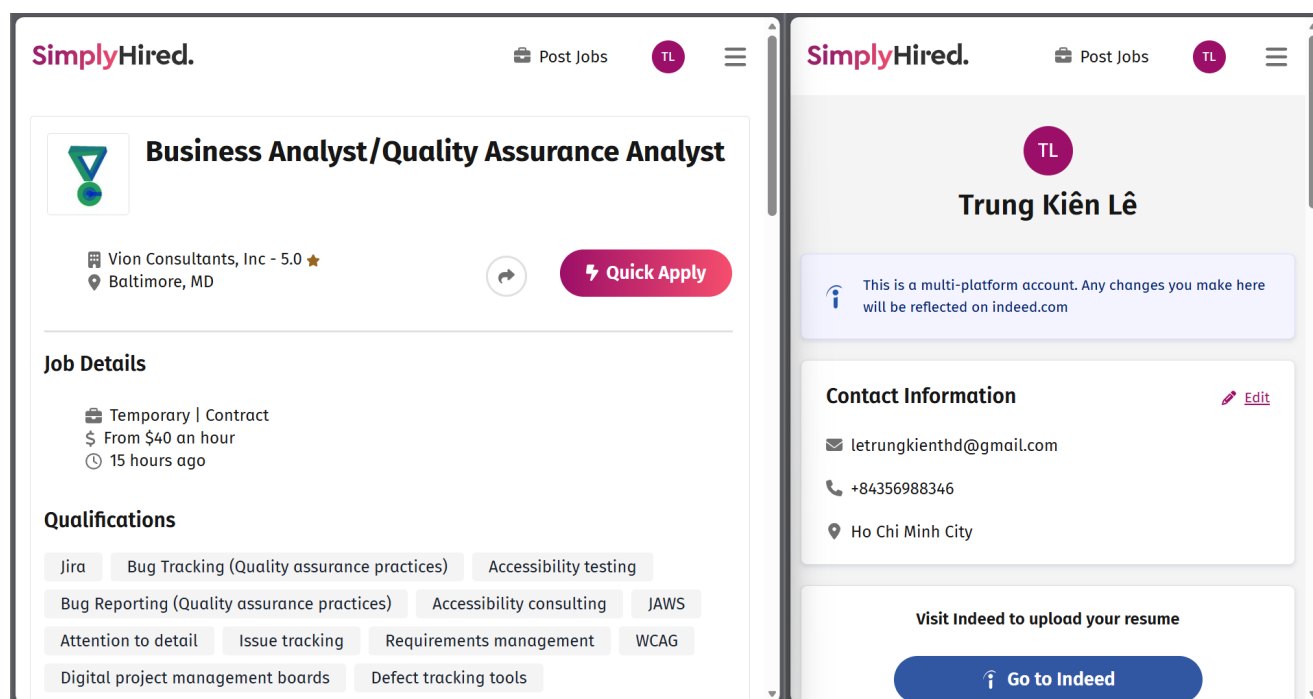
Required Skills

- Business analysis and stakeholder interviews.
- Requirements traceability across Strategic Brief, IA, and UX deliverables.
- Content audit/workbook validation, mockup acceptance criteria, and defect-log operation.
- Test planning/test cases for content migration, redirects, and site testing.
- WCAG 2.1 AA/Section 508 accessibility testing with Jira/Azure DevOps, Drupal, axe, WAVE, NVDA, and JAWS.

AI Impact Analysis (1–2 sentences)

- This role is a QA lead role for content migration, accessibility, redirects, and pre-launch verification, so AI can help draft traceability matrices, test cases, and defect summaries.
- Human review still matters because WCAG findings, content-retention decisions, and launch-risk triage require context-sensitive judgment.

Evidence



1.2.2 Posting 02

- Job title: AI Chatbot Tester
- Company: Alignerr
- Location: Vietnam (Remote)
- Platform: LinkedIn
- Posting date: 28/05/2026
- Job URL: <https://www.linkedin.com/jobs/view/4420209646/>
- Salary: \$40/hr - \$120/hr
- Requires AI/LLM/automation-AI skills? Y

Job Description

About the job

About The Role

What if your opinion could make AI smarter, safer, and more helpful for millions of people? As an AI Chatbot Tester with Alignerr, that's exactly what you'll do. We're looking for curious, thoughtful individuals to engage with cutting-edge AI chatbots, evaluate their responses, and provide the human feedback that drives real improvement in AI systems.

This is a fully remote, flexible contract role — work on your own schedule, from anywhere.

Organization: Alignerr

Type: Hourly Contract

Location: Remote

Commitment: 10–40 hours/week

What You'll Do

- Hold conversations with AI chatbots across a wide range of topics — from everyday questions to creative challenges.
- Rate responses for helpfulness, accuracy, tone, and safety.
- Identify issues such as factual errors, awkward phrasing, bias, or inappropriate outputs.
- Push the limits of AI with creative, nuanced, and challenging prompts.
- Record structured feedback that directly informs how AI systems are improved.
- Work independently and asynchronously on your own schedule.

Who You Are

- Naturally curious — you enjoy exploring ideas across different topics and asking great questions.
- Able to evaluate written responses critically and objectively.
- Clear and concise in your written communication.
- Comfortable navigating chat-based digital interfaces.
- Reliable and self-motivated when working independently.
- No technical background or prior AI experience required.

Nice to Have

- Experience in writing, editing, journalism, research, or customer service.
- Familiarity with AI tools or chatbot platforms.
- An eye for spotting subtle errors in logic, tone, or content.

Why Join Us

- Work on some of the most advanced AI projects in the world, partnering with top research labs.
- Fully remote and flexible — set your own hours and work from anywhere.
- Freelance autonomy: no micromanagement, no rigid schedules.
- Meaningful work that directly shapes the quality and safety of AI used globally.
- Potential for ongoing contracts and expanded project opportunities.

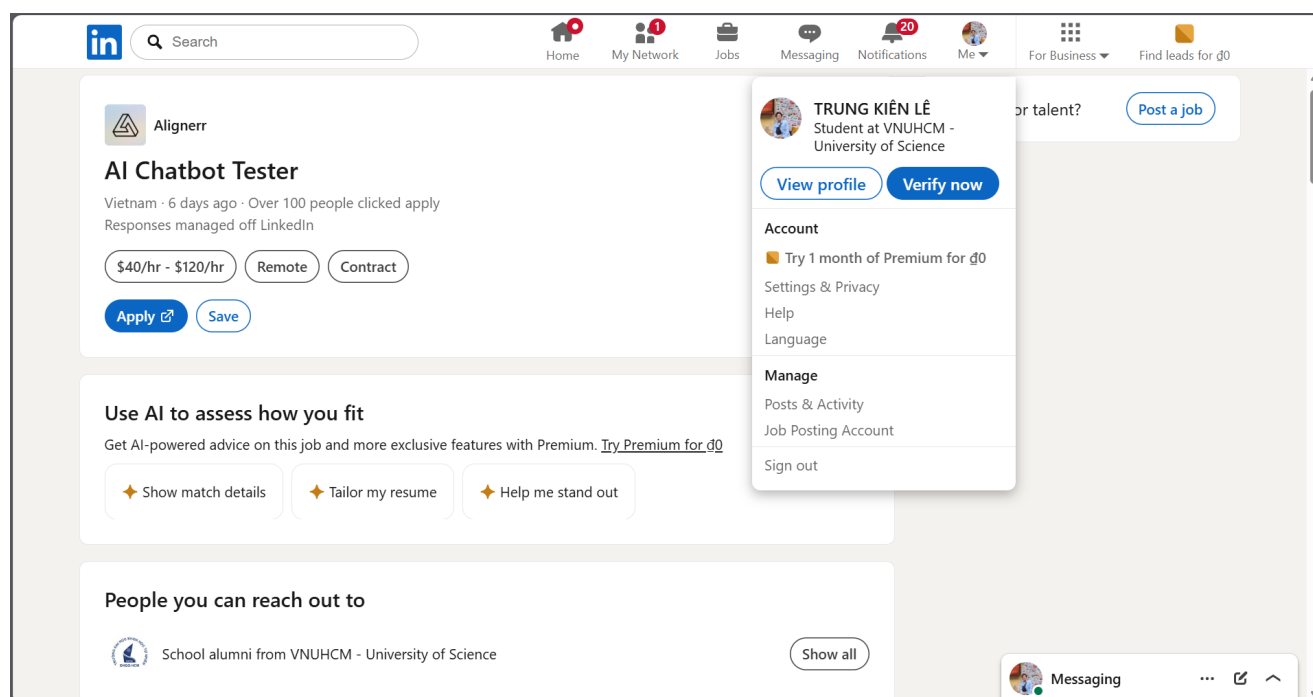
Required Skills

- AI chatbot conversation testing across broad topics.
- Response rating for helpfulness, accuracy, tone, and safety.
- Factual-error, awkward-phrasing, bias, and inappropriate-output detection.
- Adversarial/nuanced prompt design to challenge chatbot limits.
- Structured feedback writing, critical written evaluation, asynchronous remote work, and chatbot-platform familiarity.

AI Impact Analysis (1–2 sentences)

- This is a direct AI-era QA role where the product under test is LLM behavior, not only deterministic software flows.
- Human-in-the-loop testers improve model safety and usefulness by challenging chatbots, detecting subtle failures, and converting judgment into structured training feedback.

Evidence



1.2.3 Posting 03

- Job title: Fresher Full-stack Test Engineer (QA/QC/Tester)
- Company: KMS Technology, Inc.
- Location: Ho Chi Minh City, Ho Chi Minh City, Vietnam (On-site)
- Platform: LinkedIn
- Posting date: 29/05/2026
- Job URL: <https://www.linkedin.com/jobs/view/4417931013/>
- Salary: Not listed
- Requires AI/LLM/automation-AI skills? Y

Job Description

About the job

Building Products & Systems that Shape the Future

KMS Technology is a strategic engineering company helping businesses turn bold ideas into high-impact solutions—faster. Founded in 2009 as a U.S.-based services company, we've grown into a global organization with locations in the US, Vietnam, Mexico and Poland. KMS is trusted globally for the quality of our engineering and consulting services. We bring deep expertise in product development and quality assurance, Data & AI-native engineering, and delivery excellence to every engagement. Our mission is to help customers build what's next—accelerating innovation, crafting brilliant solutions, and creating real-world impact. At KMS, we believe sustainable growth is built on the success of our clients and employees and in making a lasting contribution to our communities.

Job Description

Your key responsibilities

- Develop a strong understanding of domain knowledge and client testing processes to execute testing activities effectively.
- Work closely with the project team on daily tasks and participate in sprint demo meetings with clients.
- Develop, maintain, and execute test cases/test scripts.
- Identify, report, track, and monitor defects using the defect tracking system.
- Prepare and review test documentation to ensure accuracy and completeness.
- Address issues related to testing quality and suggest improvements.
- Communicate test progress, results, and quality risks both internally and directly with the client.

Qualifications

- 4th-year student or recent graduate with a Bachelor's degree in Information Technology, Computer Science, Software Engineering, or a related field, with less than one year of experience.
- Strong IT background with a GPA of 7.5+ and academic transcript attached when submitting CV.
- Upper-intermediate or higher English proficiency, both written and verbal.
- Minimum 3-month internship experience in software testing, automation testing, or a related field is a plus.
- Strong self-learning ability with a proactive, growth-oriented mindset.
- Excellent analytical and problem-solving skills.
- Ability to work independently and collaborate effectively in a team.

Technical requirements

- Solid understanding of software testing concepts and methodologies, including manual, automation, integration, unit, and API testing.
- Solid programming skills in Python / Java / JavaScript, or other relevant languages.
- Familiarity with testing tools (e.g., Selenium, Katalon, Playwright, Appium, JUnit), as well as tools for API testing (e.g., Postman), is a plus

Nice to have

- Experience using AI chat tools (ChatGPT, Claude, Gemini, etc.) for research, debugging, and learning
- Familiarity with at least one AI coding assistant (GitHub Copilot, Cursor, Claude Code, or similar)
- Ability to write clear, contextual prompts to generate code snippets, unit tests, or documentation
- Awareness of AI output limitations and responsible AI use (data privacy, handling of sensitive client data)

Additional Information

BE YOUR BEST WITH KMS

- Working in one of the Best Places to Work in Vietnam
- Building large-scale & global software products
- Working & growing with Passionate & Talented Team
- Diverse careers opportunities with Software Outsourcing, Software Product Development, IT Solutions & Consulting
- Attractive Salary and Benefits
- Performance appraisal twice a year and performance bonus
- Onsite opportunities: short-term and long-term assignments in North American (U.S, Canada), Europe, Asia.
- Flexible working time
- Various training on hot-trend technologies, best practices, and soft skills
- Premium healthcare insurance for you and your loved ones
- Company trip, big annual year-end party every year, team building,...
- Fitness & sports activities: football, tennis, table tennis, badminton, yoga,...
- Joining community development activities: 1% Pledge, charity every quarter, blood donation, public seminars, career > - orientation talks,...
- Free in-house entertainment facilities (foosball, ping pong, gym,...), coffee (latte, cappuccino, espresso,...) and snack (instant noodles, cookies, candies,...)

And much more...Send your Resume including Academic Transcript to join us and let yourself explore other fantastic things!

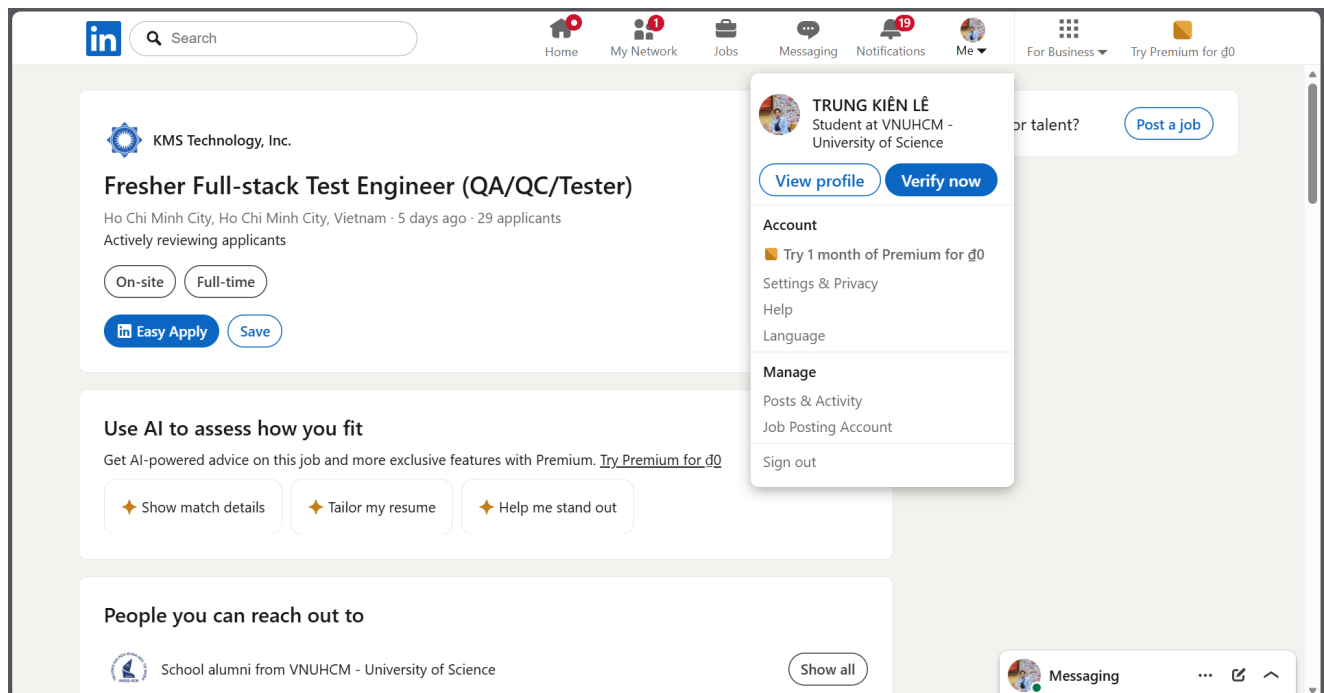
Required Skills

- Manual, automation, integration, unit, and API testing.
- Test case/test script development, defect reporting/tracking, and test documentation review.
- Python, Java, or JavaScript programming.
- Selenium, Katalon, Playwright, Appium, JUnit, and Postman familiarity.
- Sprint-demo participation and client communication.
- AI chat tools, AI coding assistants, contextual prompt writing, and responsible AI/data-privacy awareness.

AI Impact Analysis (1–2 sentences)

- This fresher role explicitly treats AI tools as productivity skills for research, debugging, learning, code snippets, unit tests, and documentation.
- It also shows employers expect junior QA engineers to understand AI limits and protect client data when using tools such as ChatGPT, Claude, Gemini, Copilot, Cursor, or Claude Code.

Evidence



1.2.4 Posting 04

- Job title: Remote QA Tester Work From Home
- Company: Apply4U | Human-Assisted AI Job & Recruitment Platform
- Location: United Kingdom (Remote)
- Platform: LinkedIn
- Posting date: 02/06/2026
- Job URL: <https://www.linkedin.com/jobs/view/4421658202/>
- Salary: £13/hr - £16/hr
- Requires AI/LLM/automation-AI skills? Y (AI-assisted platform context)

Job Description

About the job

Remote QA Tester (Manual – Entry-Level)

Location: United Kingdom (Remote)

Salary: £13 – £16 per hour

Job Type: Part-Time | Remote

About the Role

Apply4U is looking for a detail-oriented QA Tester to join our team. This is an entry-level role ideal for someone looking to start their career in software testing.

Key Responsibilities

- Test web and mobile applications for bugs and usability issues.
- Document test cases, results, and report issues clearly.
- Collaborate with developers to reproduce and resolve defects.
- Ensure applications meet quality standards before release.

What We're Looking For

- Basic knowledge of QA/testing principles (training can be provided).
- Strong attention to detail and problem-solving skills.
- Good written and verbal communication skills.
- Enthusiasm to learn and grow within the QA field.

Perks & Benefits

- Work from home with flexible hours.
- Gain valuable industry experience with real projects.
- Supportive remote team environment.

Apply Now:

- Launch your QA career with Apply4U — apply today.

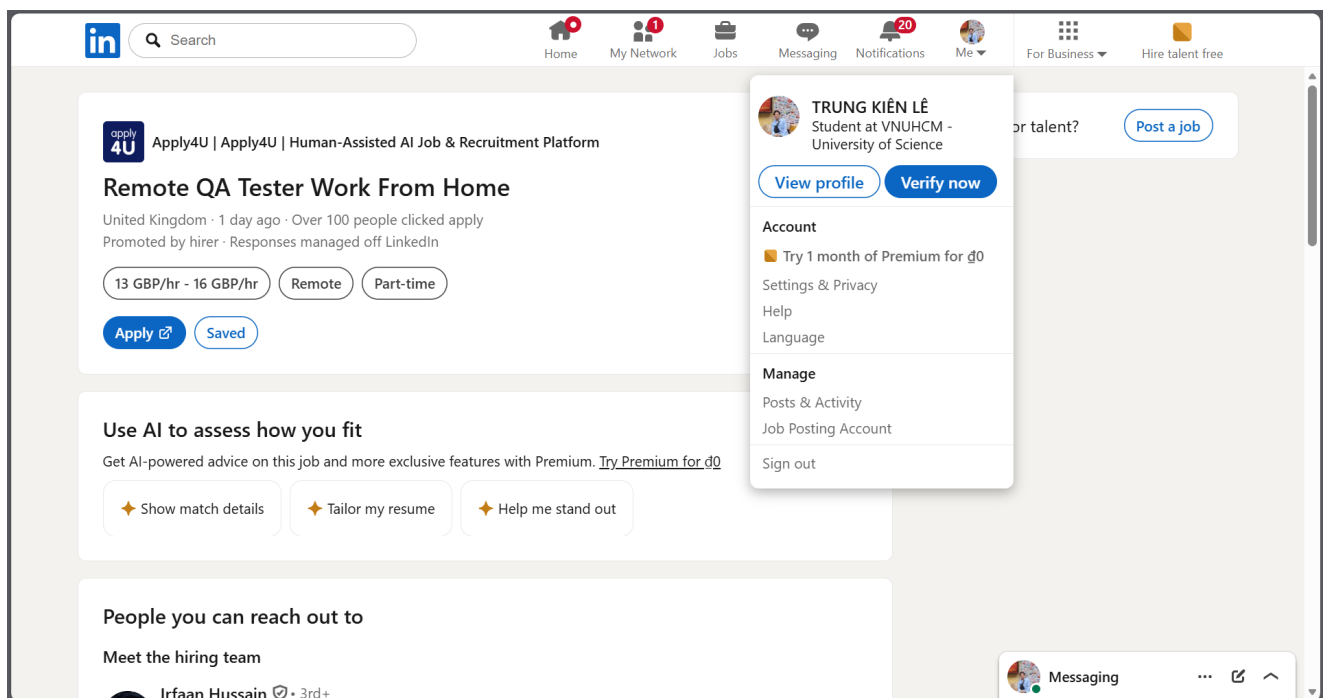
Required Skills

- Entry-level manual QA for web and mobile applications.
- Bug and usability issue identification.
- Test case/result documentation and clear issue reporting.
- Defect reproduction collaboration with developers.
- Quality-standard checks before release.
- Remote collaboration, attention to detail, problem solving, and written/verbal communication.

AI Impact Analysis (1–2 sentences)

- The posting does not require direct AI/LLM skills in the task list, but it belongs to a human-assisted AI recruitment platform where QA protects user-facing web and mobile workflows.
- AI can support repetitive test-case drafting and bug-report cleanup, while the entry-level tester still supplies human usability judgment and clear reproduction evidence.

Evidence



1.2.5 Posting 05

- Job title: Senior QA Analyst
- Company: Zwift
- Location: Long Beach, CA (Remote-eligible US locations)
- Platform: SimplyHired
- Posting date: 23/05/2026
- Job URL: https://www.simplyhired.com/job/Hna9XHBa_Kh_Knfugud6Qt2EPSUDnPF-XcQkBUodtpTum8fs5mCqxc
- Salary: \$70,500 - \$110,000 a year
- Requires AI/LLM/automation-AI skills? Y (automation/API/CI context)

Job Description

Full Job Description

Location: Remote- Eligible US Locations

About the role and about You:

Zwift seeks an experienced Sr. QA Analyst to join our team. As part of our QA Team, you will be responsible for ensuring test coverage across Zwift's family of software products. You will develop test plans, create test cases for new features and see our end-to-end testing through. You will help influence and acquire the methods, practices, and tools we'll use to drive how we test into the future.

What you'll do:

- Foster strong, communicative team spirit supporting our globally full-stack development efforts, supporting frontend and backend.

- Collaborate with cross-functional teams, including developers, product managers, and designers, to create detailed test plans and test cases based on project requirements and specifications.
- Define test strategy and lead testing from requirements gathering to release phase.
- Organize and perform regression and feature testing to reveal bugs in software.
- Craft detailed, reproducible, and concise bug reports.
- Help design, refine, and distribute QA reports and feedback to development, product managers, customer support, and senior management.
- Participate in technical design and provide a voice for testability and quality.
- Continually improve test cases to ensure coverage on all Zwift software products.
- Identify, source, define, and implement useful, specialized tools and/or methods to benefit process, automation, and reporting.

What we're looking for:

- 4+ years working in Quality Assurance focused on Mobile Application testing (iOS/Android), API testing, Server testing.
- Experience in leading testing for client-server applications and native apps.
- Strong analytical problem-solving skills.
- 2+ years of recent experience using JIRA and TestMo to support Agile development teams.
- Experience working in Agile/Scrum environments with cross-functional teams.
- Ability to design and write effective test strategies for complex systems and features.
- Deep understanding and experience in QA methodologies, tools, and practices.
- Strong analytical and problem-solving skills with the ability to operate independently.
- Database validation testing (SQL/NoSQL) to ensure data integrity and consistency.
- Experience with CI/CD tools (Jenkins, GitHub Actions, GitLab CI) and test integration.
- Bachelor's Degree in Computer Science or a related technical field, or equivalent demonstrable experience and ability.

Bonus points:

- Experience using test tools such as Appium, Postman, and JavaScript.
- Experience in testing microservices.
- Passion for cycling, running and/or fitness.
- Passion for video games and web technologies.
- Experienced Zwifter.

For All US Based Full-Time Positions:

The base salary for this position ranges between \$70,500 to \$110,000. The base salary will be based on a number of factors including the role offered, the individual's job-related knowledge, skills, qualifications, and geographic location. In addition to base salary, Zwift is proud to offer a comprehensive and competitive benefits package for all eligible employees which also includes performance bonuses, equity, and a full range of medical, financial, and other perks and benefits.

If Zwift determines in any stage of our interviews that any AI tools are being used without disclosure or citation, your candidacy will be disqualified.

How to stand out among the rest:

Your resume/CV is enough to show off your skills, accomplishments, and experience. However, if you choose to include a cover letter introducing us to your awesome personality, we will read that too.

We strongly believe that different backgrounds and ideas are a competitive advantage; we hire candidates of any race, color, ancestry, religion, sex, national origin, sexual orientation, gender identity, age, marital or family status, disability, Veteran status, and any other status. Zwift is proud to be an Equal Opportunity Employer. If you have a disability or special need that requires accommodation, please let us know by emailing careers@zwift.com.

Zwift, Inc. is an Equal Opportunity Employer.

Required Skills

- Senior QA strategy and test leadership from requirements to release.
- Mobile app testing for iOS/Android, client-server apps, native apps, APIs, and servers.
- Regression/feature testing, test plans/test cases, and reproducible bug reporting.
- QA reporting, testability review, Jira/TestMo, and Agile/Scrum collaboration.
- SQL/NoSQL data validation and CI/CD test integration.
- Appium, Postman, JavaScript, and microservices testing.

AI Impact Analysis (1–2 sentences)

- This role emphasizes senior ownership of test strategy, specialized QA tools, reporting, and future testing methods, while also warning applicants to disclose AI-tool use during interviews.
- AI may help analyze failures and draft reports, but the job signals that responsible and transparent AI use is becoming part of QA professionalism.

Evidence

The image displays two side-by-side screenshots of the SimplyHired website. The left screenshot shows a job listing for 'Senior QA Analyst' at Zwift. The job details include 'Full-time', '\$70,500 - \$110,000 a year', and '11 days ago'. A 'Quick Apply' button is visible. The qualifications listed are: Jira, Data integrity assurance, Data validation techniques, Tooling, Database testing, iOS, Smartphone applications, Agile software development, Defect tracking tools, DevOps automation, Cross-functional communication, and Test Planning (Quality assurance practices). The right screenshot shows the profile of 'Trung Kiên Lê'. It includes contact information: 'letrungkienthd@gmail.com', '+84356988346', and 'Ho Chi Minh City'. A note states: 'This is a multi-platform account. Any changes you make here will be reflected on indeed.com'. At the bottom, there is a button to 'Go to Indeed'.

1.2.6 Posting 06

- Job title: Senior Test Analyst
- Company: Cognizant
- Location: Reston, VA (Remote)
- Platform: SimplyHired
- Posting date: 01/06/2026
- Job URL:
https://www.simplyhired.com/job/Z384WERQ3uUYT3LmAcDqMvq7be7xD7HcPE8lLoBK1d45P_UUycFjSw
- Salary: \$53,477 - \$92,500 a year
- Requires AI/LLM/automation-AI skills? Y (automation/API testing)

Job Description

Full Job Description

About the role

As a Senior Test Analyst, you will make an impact by ensuring the quality, reliability, and performance of applications through robust test automation and comprehensive testing practices. You will be a valued member of the QA team and work collaboratively with developers, business stakeholders, and cross-functional Agile teams to deliver high-quality solutions.

In this role, you will:

- Develop and maintain automated test scripts using Java, Selenium WebDriver, and Rest Assured.
- Execute functional, regression, smoke, and API testing activities to ensure product quality.
- Design, create, and maintain test cases, test data, and test documentation.
- Analyze requirements to ensure adequate test coverage and identify gaps.
- Identify, log, track, and retest defects using Jira or similar tools.

Work model

We strive to provide flexibility wherever possible. Based on this role's business requirements, this is a remote position open to qualified applicants in Reston, VA. Regardless of your working arrangement, we are here to support a healthy work-life balance through our various wellbeing programs.

The working arrangements for this role are accurate as of the date of posting. This may change based on the project you're engaged in, as well as business and client requirements.

What you need to have to be considered

- 5+ years of experience in software testing and test automation.
- Hands-on experience with Java, Selenium WebDriver, and Rest Assured.
- Strong understanding of API testing, SDLC, STLC, and Agile methodologies.
- Experience with Git, Jira, and CI/CD tools such as Jenkins or GitHub Actions.
- Strong analytical, troubleshooting, and communication skills.

These will help you stand out

- Experience using GitHub Copilot or AI-assisted tools to accelerate testing productivity.
- Ability to collaborate effectively with cross-functional Agile teams.
- Strong focus on continuous quality improvement and process optimization.
- Experience in designing scalable and maintainable automation frameworks.
- Exposure to modern DevOps and test automation practices.

Salary and Other Compensation

- Applications will be accepted until June 05, 2026.
- The annual salary for this position is between \$53,477 - \$92,500 depending on experience and other qualifications of the successful candidate.
- This position is also eligible for Cognizant's discretionary annual incentive program, based on performance and subject to the terms of Cognizant's applicable plans.

Benefits

Cognizant offers the following benefits for this position, subject to applicable eligibility requirements:

- Medical/Dental/Vision/Life Insurance.
- Paid holidays plus Paid Time Off.
- 401(k) plan and contributions.
- Long-term/Short-term Disability.

Cognizant will only consider applicants for this position who are legally authorized to work in the United States without company sponsorship.

- Please note, this role is not able to offer visa transfer or sponsorship now or in the future*

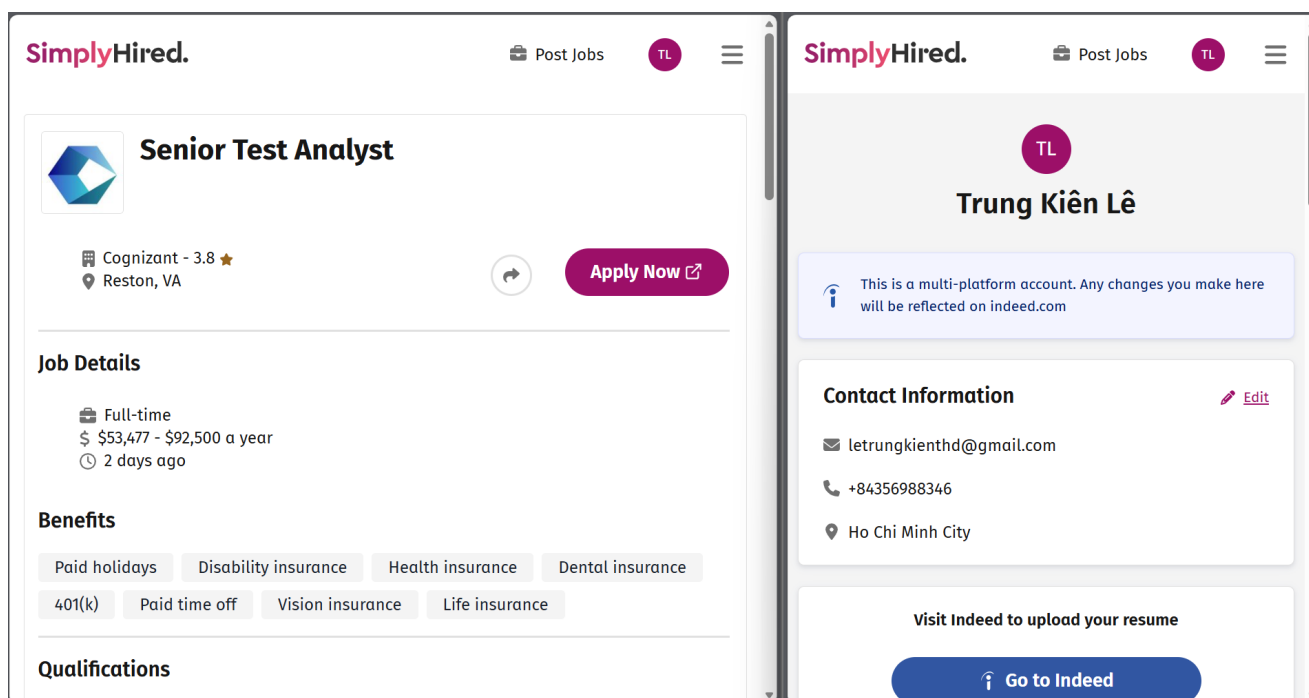
Required Skills

- 5+ years software testing and test automation.
- Java, Selenium WebDriver, and Rest Assured.
- Functional, regression, smoke, and API testing.
- Test case/test data/test documentation maintenance and requirement coverage analysis.
- Jira defect tracking/retesting, Git, Jenkins/GitHub Actions CI/CD, SDLC/STLC, and Agile collaboration.
- Scalable automation-framework design plus GitHub Copilot or AI-assisted testing tools.

AI Impact Analysis (1–2 sentences)

- This role directly names GitHub Copilot or AI-assisted tools as a differentiator for accelerating testing productivity.
- Enterprise QA is moving toward automation frameworks, API coverage, and CI/CD integration where AI can help generate scripts, find coverage gaps, and speed defect investigation.

Evidence



1.2.7 Posting 07

- Job title: Clinical Quality Assurance Analyst
- Company: Themesoft Inc
- Location: Albany, NY (Remote)
- Platform: SimplyHired
- Posting date: 20/05/2026
- Job URL:
<https://www.simplyhired.com/job/ptTM58YVddK9WzZn6iqn01UByaalq4gjIXIW73GUMQDNUcEXbxgNUg>
- Salary: \$50 - \$55 an hour
- Requires AI/LLM/automation-AI skills? N

Job Description

Full Job Description

Job Title: Clinical Quality Assurance Analyst

Location: Albany - Remote with travel

- Travel may be required up to 25% of the time.
- Some overnight stays will be required, sometimes for the entire work week.
- An expense budget will be available for travel costs.
 - Expense invoices will be due within 7 days of travel.
- Remote work will be allowed when not traveling.

Key Responsibilities:

- Evaluate child/youth's records for compliance to applicable regulations.

- Clearly communicate evaluation requirements and follow up.
- Evaluate documentation against applicable State/Federal regulations.
- Collect data and documentation to support compliance findings.
- Utilize technology to convert hard copy documentation to electronic and submit via secure file transfer.
- Record, summarize, and communicate compliance findings for the Children's Transformation Program.
- Assist in determining trends in non-compliance.
- Make appropriate and timely recommendations to project management.
- Assist with the ongoing operationalization of project processes and procedures.
- Assist with all internal and external reporting.
- Build and maintain relationships with stakeholders to provide a high level of customer service.
- Traveling to provider sites to perform case reviews.

Required Qualifications:

- Bachelor's degree in public health, healthcare management, psychology, human services, or related field.
- A minimum of 2 years of experience as a Healthcare Compliance Analyst, Auditor, Care Manager, or Medicaid Provider, in a role responsible for conducting oversight, compliance audits or chart reviews, or ensuring quality of care.
- An equivalent combination of advanced education, training, and experience will be considered.
- Experience with Medicaid fee-for-service and/or Managed Care billing.
- Proficiency in MS Excel, MS Word, MS Outlook, MS Visio, MS PowerPoint and MS SharePoint.
- Strong analytical and problem-solving abilities; able to evaluate data and make sound, evidence-based decisions.

Preferred/Desired Qualifications:

- Extensive knowledge of, and experience with, Home and Community-Based Services (HCBS) Waiver programs, including the 1915(c) NYS Children's Waiver, either as a provider or compliance officer.
- Experience working with eMedNY, CANS-NY, UAS-NY, Medicaid Health Homes, Medicaid Managed Care Organizations (MMCOs), or other related programs/systems.
- Experience with NYS Medicaid billing, including the use of various Electronic Health Records (EHR) systems.
- Experience working with Office of Mental Health (OMH), Office for People with Developmental Disabilities (OPWDD), or Office of Children and Families (OCFS).
- Ability to manage time independently while meeting deadlines and adhering to high performance standards.
- Experience with compliance audits or chart reviews.
- Ability to work with and relate to staff and demonstrate active listening skills.
- Process oriented and results-driven work strategy.
- Ability to foster teamwork with all levels of management and staff.
- Ability to work well independently and within a team.

- Exceptional customer service relationship techniques, including superior verbal and written communication skills.
- Excellent accuracy and attention to detail.

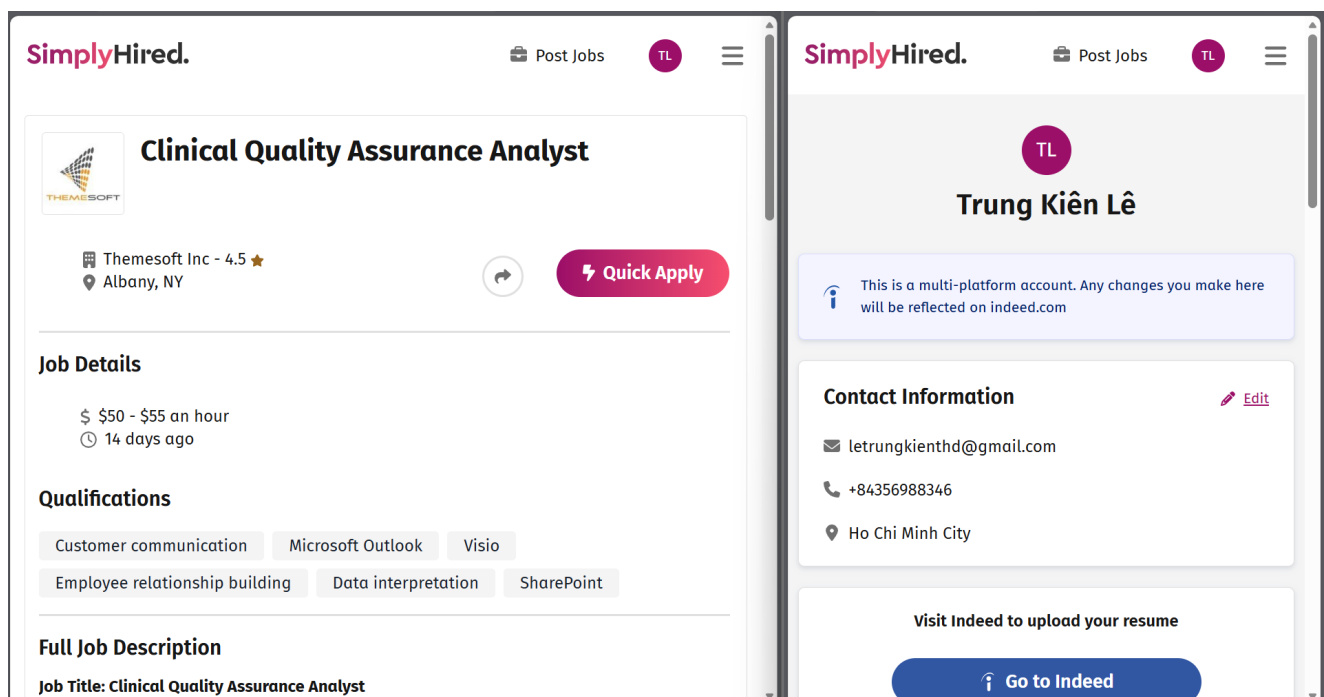
Required Skills

- Healthcare compliance audits and chart/record reviews.
- State/Federal regulation evaluation and Medicaid fee-for-service or Managed Care billing knowledge.
- Evidence collection, secure file transfer, compliance-findings documentation, and trend analysis.
- Internal/external reporting, stakeholder relationship management, and provider-site travel reviews.
- MS Excel, Word, Outlook, Visio, PowerPoint, SharePoint, and EHR/Medicaid systems exposure.

AI Impact Analysis (1–2 sentences)

- This is a non-AI clinical QA/compliance role focused on regulated healthcare documentation, Medicaid billing, and evidence-based findings.
- AI could help summarize records or detect trends, but compliance decisions, privacy handling, and provider recommendations require accountable human review.

Evidence



1.2.8 Posting 08

- Job title: Quality Assurance Specialist | \$11/hr Remote
- Company: Crossing Hurdles
- Location: Vietnam (Remote)
- Platform: LinkedIn
- Posting date: within 60-day window (promoted/current listing)
- Job URL: <https://www.linkedin.com/jobs/view/4417133986/>

- Salary: \$11/hr
- Requires AI/LLM/automation-AI skills? Y

Job Description

Full Job Description

Position: LLM - AI Quality Analyst (Personalization) - Vietnamese

Type: Short-term Contract

Compensation: \$11/Hr

Location: Remote

Commitment: 8 hours per day with 4 hours overlap with PST

Duration: 2 Months

Start Date: Immediate

Role Responsibilities

- Evaluate nuanced and ambiguous AI-generated responses with a focus on personalization quality.
- Analyze AI outputs and provide detailed quality assessments.
- Support evaluation workflows for large language model personalization tasks.
- Review and assess Vietnamese-language AI responses for accuracy, relevance, and contextual quality.
- Collaborate with global operations teams in a fast-paced remote environment.
- Contribute to improving AI quality through structured feedback and analytical insights.

Requirements

- Strong experience in data annotation, AI quality evaluation, content moderation, or related fields.
- BS/BA degree or equivalent experience in Policy, Law, Ethics, Linguistics, Journalism, Computer Science, or related analytical fields.
- Vietnamese proficiency with strong reading and writing comprehension skills.
- Exceptional analytical thinking and ability to assess nuanced AI responses.
- Willingness to use a primary personal Google account and enable personal data sources for genuine assessment tasks.
- Full-time availability in local time zone with 4 hours overlap with PST.
- Excellent communication and collaboration skills in remote work environments.

Application Process

- Apply/Easy Apply and check email for application form.
- Fill Google form.
- Complete the assessment link after shortlisting within 24 hours.

Required Skills

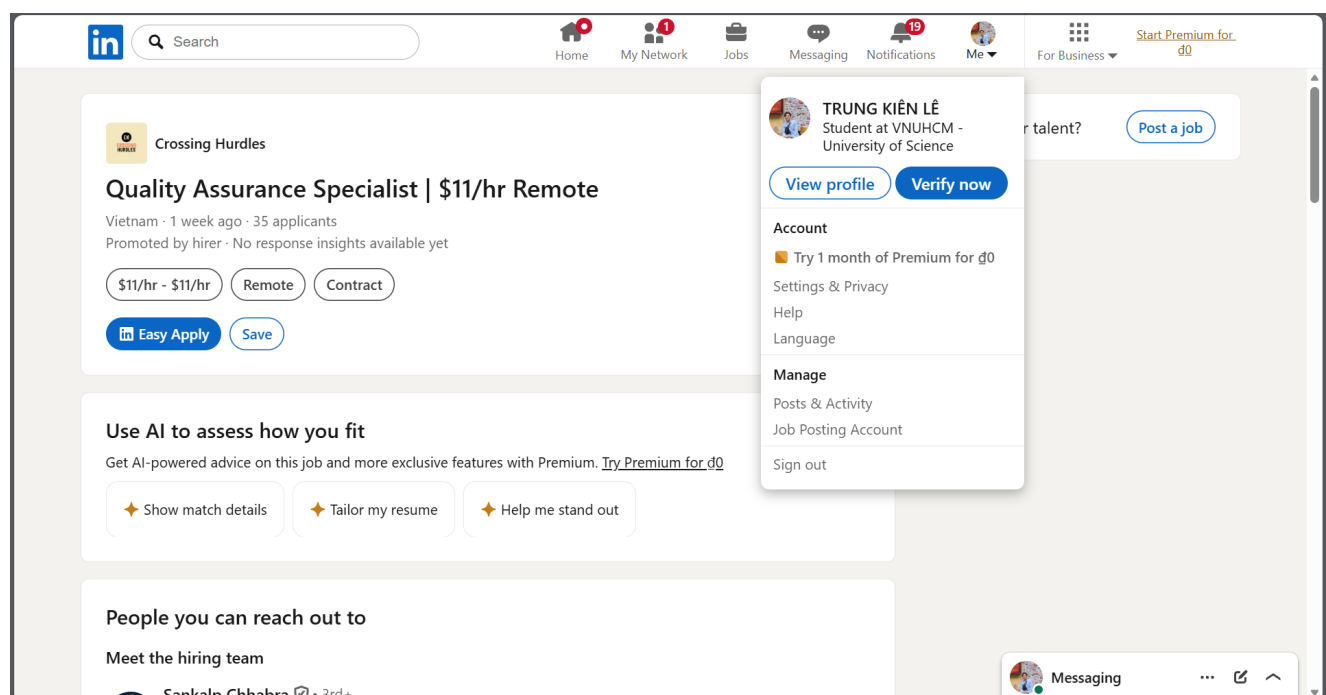
- LLM/AI quality evaluation and personalization-quality assessment.
- Nuanced and ambiguous AI-response review.

- Vietnamese-language accuracy, relevance, and context evaluation.
- Data annotation or content moderation experience.
- Structured quality feedback, analytical thinking, and remote collaboration with global operations teams.
- Policy, law, ethics, linguistics, journalism, or CS background plus personal-data-source assessment workflow awareness.

AI Impact Analysis (1–2 sentences)

- This role is built around LLM personalization QA, where testers judge whether AI responses fit Vietnamese-language context, user relevance, and quality expectations.
- It shows AI QA jobs need language expertise, ethical judgment, and careful handling of personal data sources, not only technical test automation.

Evidence



1.2.9 Posting 09

- Job title: QA Engineer – Hybrid HCM (Experience in ML/Data or Security Testing)
- Company: Dwarves Foundation
- Location: Ho Chi Minh City, Vietnam (Hybrid)
- Platform: LinkedIn
- Posting date: 1 week ago
- Job URL: <https://www.linkedin.com/jobs/view/4419144549/>
- Salary: \$1,800 - \$2,000
- Requires AI/LLM/automation-AI skills? Y (ML/data testing scope)

Job Description

About the job

Position: QA Engineer

Location: Hybrid – Ho Chi Minh City

Working time: 09:00 AM – 06:00 PM (Monday – Friday)

Salary: \$1,800 – \$2,000

🔗 About the Role: This role focuses on automation testing across API, backend, and UI, with additional scope in ML features, security/authentication testing, and API partnerships/integration with client interactions

Responsibilities

- Design, develop, and maintain automated test frameworks for API, backend, and UI layers.
- Implement and execute automated test scripts to ensure product quality and reliability.
- Collaborate closely with Engineering, Product, and DevOps teams throughout the SDLC.
- Apply Shift-Left Testing practices to identify defects early.
- Integrate automated tests into CI/CD pipelines.
- Perform test planning, execution, and defect tracking.
- Analyze test results and provide clear quality insights and recommendations.
- Contribute to testing standards, best practices, and continuous improvement initiatives.
- Support junior QEs through technical guidance and knowledge sharing.
- Participate in code reviews and quality discussions.

Additional Responsibilities

- Validate Machine Learning (ML) related features and data outputs.
- Perform testing related to security, authentication, and authorization mechanisms.
- Test and validate API partnerships and third-party integrations with client systems.
- Ensure reliability and security of client-facing API interactions.

Requirements

- 5+ years of experience in Quality Control / Test Automation.
- Strong experience in automated testing across API, backend, and UI.
- Hands-on experience with Playwright, Cypress, RestAssured, and Postman.
- Proficiency in one or more programming languages: Golang, TypeScript, Ruby, Python, or Java.
- Solid understanding of software testing methodologies and SDLC.
- Experience integrating automated tests into CI/CD pipelines.
- Familiarity with Docker and containerized test environments.
- Strong analytical and communication skills.
- Good English communication skills.

Preferred Qualifications

- Experience testing Machine Learning (ML) workflows or ML-related features.
- Knowledge of security testing, including authentication, authorization, and encryption.
- Experience validating API integrations or third-party services.
- Experience working in microservices-based architectures.
- Familiarity with Kubernetes.

Candidate Profile

- Strong hands-on automation engineer with senior-level ownership
- Quality-driven mindset with high attention to detail
- Comfortable working in fast-paced engineering environments

Learn more about us:

🌐 Website: <https://d.foundation>

🔗 LinkedIn: <https://www.linkedin.com/company/dwarvesf/>

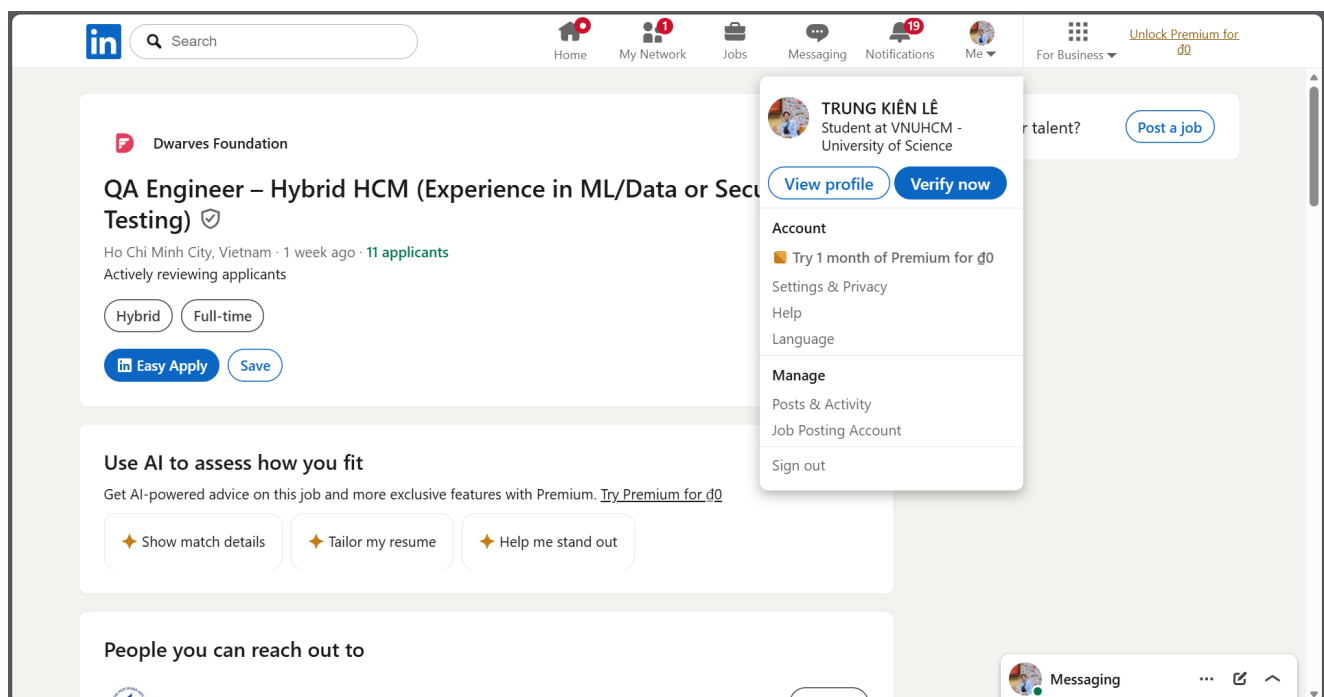
Required Skills

- Senior test automation and API/backend/UI automation-framework design.
- Automated test script implementation with Playwright, Cypress, RestAssured, and Postman.
- Golang, TypeScript, Ruby, Python, or Java programming.
- SDLC, Shift-Left Testing, CI/CD pipeline integration, Docker, and containerized test environments.
- Defect tracking, quality insights, code review participation, microservices, and Kubernetes exposure.
- ML feature/data-output validation, security/authentication/authorization/encryption testing, and API partnership/third-party integration testing.

AI Impact Analysis (1–2 sentences)

- This role shows QA expanding into ML-related feature validation, where correctness includes model/data outputs in addition to normal API, backend, and UI behavior.
- It also links AI-era QA with security, authentication, and third-party API reliability because ML features often depend on integrated data pipelines and client-facing services.

Evidence



1.2.10 Posting 10

- Job title: Audio Quality Assurance Specialist | \$50/hr Remote
- Company: Crossing Hurdles
- Location: United States (Remote)
- Platform: LinkedIn
- Posting date: 1 week ago
- Job URL: <https://www.linkedin.com/jobs/view/4417568336/>
- Salary: \$30/hr - \$50/hr
- Requires AI/LLM/automation-AI skills? Y

Job Description

About the job

Position: Audio Engineer

Type: Contract

Compensation: \$30 - \$50/hour

Location: Remote

Commitment: 10-40 hrs/week

Role Responsibilities

- Conduct detailed quality assurance on audio recordings by identifying and documenting artifacts or anomalies.
- Assess the severity of audio issues and classify them accurately for AI model training use.
- Perform audio cleanup and restoration tasks to ensure data meets professional standards.
- Annotate audio and video data with clear, concise notes to support AI/ML development.
- Participate in labeling, review, and quality control processes for large-scale datasets.
- Collaborate effectively with a distributed team, providing regular feedback and contributing to a culture of high standards.

Requirements

- Have a professional monitoring setup, including studio monitors or high-end headphones, suitable for critical listening tasks.
- Possess a strong command of written and verbal English communication for precise issue reporting and team collaboration.
- Have a deep understanding of audio quality assurance processes and artifact detection methodologies.
- Have experience with audio restoration, cleanup, and editing workflows using industry-standard tools.
- Be self-motivated, detail-oriented, and comfortable working independently in a remote environment.

Application Process

- Easy Apply on LinkedIn.

- Check email for next steps.
- Participate in resume evaluation and interview stage.

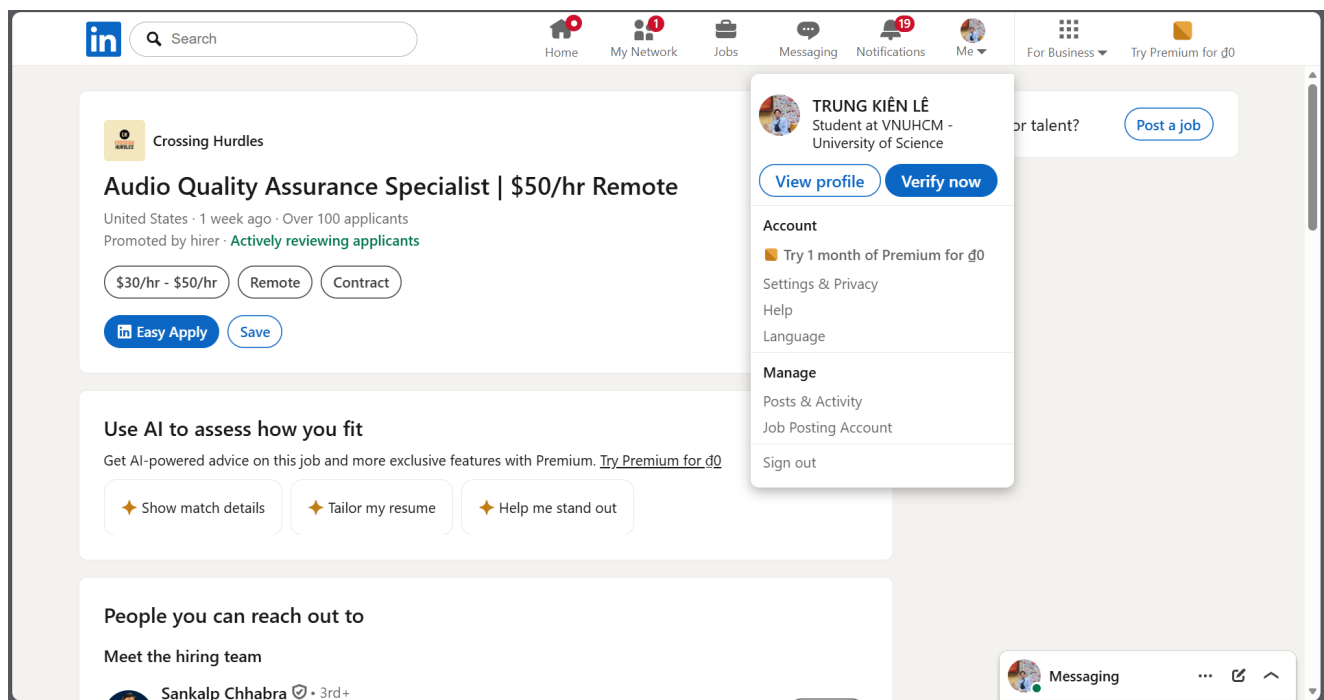
Required Skills

- Audio quality assurance and critical listening with studio monitors/high-end headphones.
- Artifact/anomaly detection and severity classification.
- Audio cleanup, restoration, and editing workflows.
- Data annotation for audio/video and labeling/review/quality-control processes for large datasets.
- Concise issue notes for AI/ML development, distributed-team collaboration, written/verbal English communication, and independent remote work.

AI Impact Analysis (1–2 sentences)

- This role shows multimodal AI creating specialized QA demand for audio datasets used in model training.
- The tester's domain expertise turns subjective audio defects into labeled, severity-classified feedback that improves AI/ML data quality.

Evidence



2) Requirement 2 — 20 Software Defects (2022–2026) (20 pts)

2.1 Summary

- Total defects: 20/20
- AI/LLM-related defects: 7 (must be ≥ 5)
- AI bias/hallucination analysis entries: 20/20 (one per defect)

2.2 20 Software Defects

Part 1: AI & Large Language Model (LLM) Defects

1. CVE-2023-29374 (LangChain Prompt Injection)

- **Source Link:** [CVE-2023-29374](#)
- **Description:** LangChain through version 0.0.131 allows prompt injection attacks against the `LLMMathChain` chain. A crafted prompt can execute arbitrary code through the Python `exec` method.
- **Severity:** Critical (CVSS 9.8)
- **Consequences:** Remote code execution on the server running the vulnerable LangChain application when untrusted prompts reach the affected chain.
- **Solution:** Upgrade LangChain to version 0.0.132 or later and avoid sending untrusted prompts to components that can execute generated code.
- **AI Hallucination/Bias:** The source names Python `exec`, not `numexpr.evaluate()`, as the execution path.

2. CVE-2024-37032 (Ollama Path Traversal)

- **Source Link:** [CVE-2024-37032](#)
- **Description:** Ollama before version 0.1.34 does not validate digest format when building model blob paths. Invalid digest values, including an initial `../` substring, can be mishandled.
- **Severity:** High (CVSS 8.8)
- **Consequences:** Path traversal through crafted digest values during model path handling. Impact should be described as filesystem path mishandling, not guaranteed full system takeover.
- **Solution:** Update Ollama to version 0.1.34 or later.
- **AI Hallucination/Bias:** Critical 9.8 and "full system takeover" overstated the source-backed severity and impact.

3. CVE-2023-39325 (Go HTTP/2 Rapid Reset DoS)

- **Source Link:** [CVE-2023-39325](#)
- **Description:** A malicious HTTP/2 client can rapidly create requests and immediately reset them, causing excessive server resource consumption in Go HTTP/2 servers.
- **Severity:** High (CVSS 7.5)
- **Consequences:** Denial of service through excessive handler goroutine/resource consumption, even though the total number of concurrent streams is bounded.
- **Solution:** Apply patched Go and golang.org/x/net/http2 versions that bound simultaneously executing handler goroutines to the stream concurrency limit.
- **AI Hallucination/Bias:** The source is Go HTTP/2 rapid reset DoS, not Anyscale Ray / ShadowRay.

4. CVE-2024-24590 (ClearML Unsafe Deserialization)

- **Source Link:** [CVE-2024-24590](#)
- **Description:** Allegro AI ClearML client SDK versions 0.17.0 through 1.14.2 can deserialize untrusted data. A maliciously uploaded artifact can run arbitrary code on an end user's system when interacted with.
- **Severity:** High (CVSS 8.0)

- **Consequences:** Arbitrary code execution on the user's system when the malicious artifact is opened or otherwise interacted with, not automatic execution on every worker preprocessing event.
- **Solution:** Upgrade ClearML client SDK to version 1.14.3 or later.
- **AI Hallucination/Bias:** The source severity is High and affected versions include 1.14.2, so 1.14.2 is not the fixed version.

5. CVE-2024-21515 (OpenCart Reflected XSS)

- **Source Link:** [CVE-2024-21515](#)
- **Description:** OpenCart versions from 4.0.0.0 have a reflected XSS issue in the `filename` parameter of the admin `tool/log` route.
- **Severity:** Medium (CVSS 4.2)
- **Consequences:** An attacker can trick a user into opening a crafted URL and steal the user's token. If the victim is an administrator, this can become a starting point for chained admin-functionality exploits.
- **Solution:** Apply OpenCart fixes for the admin redirect/XSS issue, avoid using malicious links while authenticated, and restrict or rename the admin directory as recommended by the source notes.
- **AI Hallucination/Bias:** The source describes OpenCart reflected XSS, not Flowise LLM RCE.

6. CVE-2024-22419 (Vyper concat Buffer Overwrite)

- **Source Link:** [CVE-2024-22419](#)
- **Description:** Vyper's `concat` built-in can write past the bounds of its allocated memory buffer because `build_IR` for `concat` does not properly follow the copy-function API.
- **Severity:** High (CVSS 7.3)
- **Consequences:** Valid contract data can be overwritten, changing smart-contract semantics in length-dependent cases that may not appear during normal testing.
- **Solution:** Upgrade Vyper to version 0.4.0 or later.
- **AI Hallucination/Bias:** The source is a Vyper smart-contract compiler issue, not a ComfyUI workflow-execution flaw.

7. CVE-2024-34359 (llama-cpp-python Jinja2 SSTI)

- **Source Link:** [CVE-2024-34359](#)
- **Description:** `llama-cpp-python` versions 0.2.30 through 0.2.71 load chat-template metadata from `.gguf` models and parse it with a sandbox-less Jinja2 environment.
- **Severity:** Critical (CVSS 9.7)
- **Consequences:** A crafted model chat template can trigger Jinja2 server-side template injection and lead to remote code execution when the template is rendered.
- **Solution:** Upgrade `llama-cpp-python` to a patched version outside the affected 0.2.30 through 0.2.71 range and avoid loading untrusted model files.
- **AI Hallucination/Bias:** The source describes llama-cpp-python Jinja2 SSTI/RCE, not LlamaIndex SSRF.

Part 2: Core Infrastructure & Enterprise Software Defects

8. CVE-2024-3094 (XZ Utils Malicious Backdoor)

- **Source Link:** [CVE-2024-3094](#)
- **Description:** Malicious code was deliberately injected into upstream XZ Utils tarballs starting with version 5.6.0. The obfuscated build process modified `liblzma` functions used by linked software.
- **Severity:** Critical (CVSS 10.0)
- **Consequences:** On affected builds, software linked against the modified `liblzma` could have its data interactions intercepted and modified. Practical OpenSSH exploitation depended on specific build conditions and attacker-controlled trigger material.
- **Solution:** Immediately downgrade XZ Utils to a safe version such as 5.4.6 or install distribution-specific vendor security updates.
- **AI Hallucination/Bias:** The original consequence was too broad because exploitation depended on affected builds and a matching trigger/private key.

9. CVE-2024-6387 (OpenSSH "regreSSHion" Race Condition)

- **Source Link:** [CVE-2024-6387](#)
- **Description:** A regression of CVE-2006-5051 in OpenSSH server (`sshd`) introduces a race condition where some signals can be handled unsafely.
- **Severity:** High (CVSS 8.1)
- **Consequences:** An unauthenticated remote attacker may be able to trigger the race by failing to authenticate within the grace period, with possible code execution on vulnerable glibc-based Linux targets.
- **Solution:** Upgrade OpenSSH to version 9.8p1 or set `LoginGraceTime 0` as a temporary workaround.
- **AI Hallucination/Bias:** Root execution should be phrased as possible and timing-dependent, not guaranteed.

10. CVE-2022-22965 (Spring4Shell)

- **Source Link:** [CVE-2022-22965](#)
- **Description:** A Spring MVC or Spring WebFlux application running on JDK 9+ may be vulnerable to remote code execution through data binding.
- **Severity:** Critical (CVSS 9.8)
- **Consequences:** The widely known exploit requires Tomcat as a WAR deployment. Spring Boot executable JAR deployments are not vulnerable to that exploit path, though the underlying vulnerability is more general.
- **Solution:** Upgrade Spring Framework to versions 5.3.18 / 5.2.20 or higher.
- **AI Hallucination/Bias:** The source requires specific JDK 9+ and Tomcat WAR deployment conditions for the common exploit.

11. CVE-2023-38831 (WinRAR Arbitrary Execution)

- **Source Link:** [CVE-2023-38831](#)
- **Description:** WinRAR before version 6.23 can execute arbitrary code when a user attempts to view a benign file inside a crafted ZIP archive that also contains a same-named folder with executable content.
- **Severity:** High (CVSS 7.8)
- **Consequences:** Arbitrary code execution occurs after user interaction with the crafted archive; it is not a purely silent, no-click server-side compromise.

- **Solution:** Upgrade WinRAR to version 6.23 or newer.
- **AI Hallucination/Bias:** The exploit still depends on user interaction with a crafted archive.

12. CVE-2023-4863 (libwebp Heap Buffer Overflow)

- **Source Link:** [CVE-2023-4863](#)
- **Description:** A heap buffer overflow in `libwebp` in Google Chrome before 116.0.5845.187 and `libwebp` before 1.3.2 allows out-of-bounds memory write through crafted WebP content.
- **Severity:** High (CVSS 8.8; Chromium severity Critical)
- **Consequences:** Memory corruption and possible remote code execution when crafted WebP content is processed by a vulnerable browser or application.
- **Solution:** Update Chrome, Firefox, Electron apps, or other parent software to versions that include the patched `libwebp` 1.3.2 or later.
- **AI Hallucination/Bias:** CVSS 8.8 maps to High even though Chromium separately rated the issue Critical.

13. CVE-2024-21626 (runc Container Escape)

- **Source Link:** [CVE-2024-21626](#)
- **Description:** `runc` 1.1.11 and earlier leak an internal file descriptor, allowing a newly spawned container process to have a working directory in the host filesystem namespace.
- **Severity:** High (CVSS 8.6)
- **Consequences:** A malicious image or `runc exec` workflow can gain host filesystem access. Some variants can overwrite semi-arbitrary host binaries and produce complete container escape.
- **Solution:** Update `runc` to version 1.1.12 or higher.
- **AI Hallucination/Bias:** CVSS 8.6 is High, and complete escape applies to specific variants rather than every exploit path.

14. CVE-2024-3400 (Palo Alto Networks PAN-OS Injection)

- **Source Link:** [CVE-2024-3400](#)
- **Description:** A command injection issue from arbitrary file creation affects the GlobalProtect feature in specific PAN-OS versions and feature configurations.
- **Severity:** Critical (CVSS 10.0)
- **Consequences:** An unauthenticated attacker can execute arbitrary code with root privileges on vulnerable firewall deployments. Cloud NGFW, Panorama appliances, and Prisma Access are not impacted by this CVE.
- **Solution:** Apply Palo Alto Networks hotfixes for affected PAN-OS 10.2, 11.0, and 11.1 releases and ensure vulnerable GlobalProtect gateway configurations are not exposed.
- **AI Hallucination/Bias:** The source limits risk to specific PAN-OS versions and GlobalProtect feature configurations.

15. CVE-2022-42889 (Text4Shell)

- **Source Link:** [CVE-2022-42889](#)
- **Description:** Apache Commons Text versions 1.5 through 1.9 include default string lookups such as `script`, `dns`, and `url` in variable interpolation.
- **Severity:** Critical (CVSS 9.8)

- **Consequences:** Applications may execute code or contact remote servers if untrusted configuration values reach the vulnerable interpolation defaults.
- **Solution:** Upgrade Apache Commons Text to version 1.10.0 or higher, which disables the problematic interpolators by default.
- **AI Hallucination/Bias:** The impact depends on attacker-controlled input reaching vulnerable interpolation.

16. CVE-2023-22515 (Atlassian Confluence Privilege Escalation)

- **Source Link:** [CVE-2023-22515](#)
- **Description:** Publicly accessible Atlassian Confluence Data Center and Server instances could be exploited by external attackers to create unauthorized Confluence administrator accounts.
- **Severity:** Critical (CVSS 10.0)
- **Consequences:** Attackers can create administrator accounts and access or take over affected Confluence instances. Atlassian Cloud sites are not affected.
- **Solution:** Upgrade to fixed Confluence versions such as 8.3.3, 8.4.3, 8.5.2, or newer fixed releases.
- **AI Hallucination/Bias:** "Broken object-level authorization" was imprecise; the source describes unauthenticated administrator-account creation.

17. CVE-2024-2961 (glibc iconv Buffer Overflow)

- **Source Link:** [CVE-2024-2961](#)
- **Description:** The `iconv()` function in GNU C Library 2.39 and older can overflow the output buffer by up to 4 bytes when converting to the ISO-2022-CN-EXT character set.
- **Severity:** High (CVSS 8.1)
- **Consequences:** The official impact is application crash or neighboring-variable overwrite. Remote code execution requires an additional exploit chain and should not be presented as direct default impact.
- **Solution:** Apply the upstream glibc patch or install updated Linux distribution packages.
- **AI Hallucination/Bias:** RCE needs careful conditional wording because the source names crash / neighboring-variable overwrite.

18. CVE-2023-50164 (Apache Struts 2 File Upload Exploit)

- **Source Link:** [CVE-2023-50164](#)
- **Description:** Apache Struts allows attackers to manipulate file-upload parameters to enable path traversal during multipart uploads.
- **Severity:** Critical (CVSS 9.8)
- **Consequences:** Under some circumstances, an attacker can upload a malicious file that can be used for remote code execution. The exploitability depends on deployment and upload handling.
- **Solution:** Update Apache Struts to versions 2.5.33, 6.3.0.2, or later.
- **AI Hallucination/Bias:** RCE is source-backed but conditional, not guaranteed for every upload path.

19. CVE-2022-26134 (Atlassian Confluence OGNL Injection)

- **Source Link:** [CVE-2022-26134](#)

- **Description:** Affected Confluence Server and Data Center versions contain an OGNL injection vulnerability that allows unauthenticated attackers to execute arbitrary code.
- **Severity:** Critical (CVSS 9.8)
- **Consequences:** Complete unauthenticated remote code execution on vulnerable Confluence Server or Data Center instances.
- **Solution:** Upgrade Confluence to fixed versions such as 7.4.17, 7.13.7, 7.14.3, 7.15.2, 7.16.4, 7.17.4, 7.18.1, or later supported fixed releases.
- **AI Hallucination/Bias:** "Structural code parsing from HTTP request headers" was unclear; the source says unauthenticated OGNL injection.

20. CVE-2024-47575 (Fortinet FortiManager "FortiJump")

- **Source Link:** [CVE-2024-47575](#)
- **Description:** Fortinet FortiManager and FortiManager Cloud contain a missing-authentication vulnerability for a critical function across multiple 7.x and 6.x branches.
- **Severity:** Critical (CVSS 9.8)
- **Consequences:** An unauthenticated attacker can execute arbitrary code or commands through specially crafted requests.
- **Solution:** Upgrade to fixed FortiManager/FortiManager Cloud branches, including 7.6.1, 7.4.5, 7.2.8, 7.0.13, 6.4.15, 6.2.13, or vendor-designated Cloud fixed versions as applicable.
- **AI Hallucination/Bias:** The source impact is arbitrary code/command execution, and the fix list includes older affected branches.

3) Requirement 3 — Test Cases for ONE Physical Product (40 pts)

3.1 Device Declaration

- Device type: Domestic stand fan with electronic/soft-button control panel
- Brand: Lifan
- Model: LIFAN D-16RC-0
- Year: Don't remember
- Serial number (masked middle 4 chars): Don't remember

3.2 Required Device Evidence



3.3 Test Cases (15 total)

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
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TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC01	[Functional] Verify that the ON/OFF button powers on the fan from standby/off state.	Fan is assembled safely, connected to rated AC supply, and currently OFF. Control panel is visible.	1. Press the ON/OFF button once. 2. Observe fan operation and panel indicators for 5 seconds.	Fan changes to ON state. Fan motor runs and produces airflow. Control panel shows active power status. No other function requires manual activation to start basic operation.	The fan is still OFF	Failed
TC02	[Functional] Verify that the ON/OFF button powers off the fan during operation.	Fan is ON and producing airflow at any valid speed.	1. Press the ON/OFF button once. 2. Observe motor, airflow, swing state, and panel indicators for 5 seconds.	Fan changes to OFF state. Motor stops, airflow stops, oscillation stops if active, and control panel no longer indicates active operation except any allowed standby indication.	The fan is still ON	Failed
TC03	[Functional] Verify SPEED cycles from Low to Medium.	Fan is ON. Current speed indicator is Low.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from Low to Medium. Medium speed indicator is active. Airflow becomes stronger than Low.	Same as expected result	Passed
TC04	[Functional] Verify SPEED cycles from Medium to High.	Fan is ON. Current speed indicator is Medium.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from Medium to High. High speed indicator is active. Airflow becomes stronger than Medium.	Same as expected result	Passed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC05	[Functional / Boundary] Verify SPEED wraps from High back to Low.	Fan is ON. Current speed indicator is High.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from High to Low. Low speed indicator is active. Airflow returns to the lowest available intensity. No undefined speed state appears.	Same as expected result	Passed
TC06	[Functional] Verify MODE cycles from Normal to Natural.	Fan is ON. Current mode indicator is Normal.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Normal to Natural. Natural mode indicator is active. Fan remains ON and uses the Natural wind profile.	The MODE button did not function	Failed
TC07	[Functional] Verify MODE cycles from Natural to Sleep.	Fan is ON. Current mode indicator is Natural.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Natural to Sleep. Sleep mode indicator is active. Fan remains ON and uses the Sleep wind profile.	The MODE button did not function	Failed
TC08	[Functional / Boundary] Verify MODE wraps from Sleep back to Normal.	Fan is ON. Current mode indicator is Sleep.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Sleep to Normal. Normal mode indicator is active. No undefined wind profile appears.	The MODE button did not function	Failed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC09	[Functional] Verify SWING enables horizontal oscillation.	Fan is ON. Swing is currently OFF. Fan head has safe clearance for oscillation.	1. Press SWING once. 2. Observe fan head movement and swing indicator for 10 seconds.	Horizontal oscillation starts. Swing indicator shows ON state. Fan continues producing airflow at the selected speed and mode.	Same as expected result	Passed
TC10	[Functional] Verify SWING disables horizontal oscillation.	Fan is ON. Swing is currently ON and horizontal oscillation is visible.	1. Press SWING once. 2. Observe fan head movement and swing indicator for 10 seconds.	Horizontal oscillation stops. Swing indicator shows OFF state. Fan continues producing airflow at the selected speed and mode.	Same as expected result	Passed
TC11	[Functional] Verify TIMER increments from no timer to the first supported timer interval.	Fan is ON. Timer is currently OFF or no timer indicator is active.	1. Press TIMER once. 2. Observe timer indicator immediately after the press.	Timer changes from OFF to the first supported auto-off interval. The corresponding timer indicator is active. Fan remains ON.	The TIMER button did not function	Failed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC12	[Functional / Boundary] Verify TIMER cycles through all supported intervals and returns to no-timer state.	Fan is ON. Timer indicator is visible. Manufacturer-supported timer interval markings are known from the panel/manual.	1. Press TIMER repeatedly one step at a time. 2. After each press, record the active timer indicator. 3. Continue until the sequence passes the maximum supported interval.	Timer indicator advances exactly one interval per press. After the maximum supported interval, the next TIMER press returns to no-timer/OFF timer state. No skipped, duplicated, or undefined timer state appears.	The MODE button did not function	Failed
TC13	[Functional] Verify that the fan defaults to Low speed when powered ON from the standby/OFF state.	Fan is connected to rated AC supply and currently OFF. Control panel is clear and visible.	1. Press the ON/OFF button once to power ON the fan. 2. Observe the active speed indicator and initial airflow intensity.	Fan powers ON successfully. The fan speed automatically defaults to Low. The Low speed indicator is illuminated, and the initial airflow matches the Low setting.	The default speed is Medium, then change to Low	Failed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC14	[Safety] Verify that the fan guard (grille) mesh layout prevents human fingers from making contact with the rotating blades.	Fan is fully assembled, powered ON, and the blades are rotating at High speed. A standard test finger or safe visual inspection tool is available.	1. Gently attempt to pass a safe finger-sized probe through the front and rear fan guards at various gap locations. 2. Verify if the probe can reach the rotating blades.	The fan guard gaps are narrow enough to block the probe entirely. It is physically impossible for a finger to touch or come into dangerous proximity with the rotating fan blades from any angle.	Same as expected result	Passed
TC15	[Appearance / UI] Verify the legibility, print quality, and clarity of all text and markings on the fan body.	Fan is placed in a well-lit test environment. Tester inspects the device from a standard user distance.	1. Read and inspect all text on the fan body (Brand logo "LIFAN", Model "D-16RC-0", and button labels: SPEED, TIMER, SWING, MODE, ON/OFF). 2. Check for printing defects, fading, or misspellings.	All text, symbols, and labels are perfectly legible, sharply printed, and easy to read. Font alignment is consistent, and there are no signs of smudging, faded ink, or spelling errors.	Same as expected result	Passed

3.4 Test Execution Video Log (>=5)

Video ID	TC ID	Title	Platform	Link
VID-01	TC-01	[23127075 _ SOFTWARE TESTING] TC 01	YouTube Unlisted	https://www.youtube.com/watch?v=xYFowoA903U

Video ID	TC ID	Title	Platform	Link
VID-02	TC-02	[23127075 _ SOFTWARE TESTING] TC 02	YouTube Unlisted	https://www.youtube.com/watch?v=4-cxZAMtfmg
VID-03	TC-03	[23127075 _ SOFTWARE TESTING] TC 03	YouTube Unlisted	https://www.youtube.com/watch?v=EjQsOBYgneg
VID-04	TC-13	[23127075 _ SOFTWARE TESTING] TC 13	YouTube Unlisted	https://www.youtube.com/watch?v=zyAoDVczHws
VID-05	TC-14	[23127075 _ SOFTWARE TESTING] TC 14	YouTube Unlisted	https://www.youtube.com/watch?v=stYr6CmpUu8

3.5 AI-missed Edge Cases (>=3 mandatory)

AI conversation screenshot proving AI did not generate it

TC13	[Functional / Boundary] Verify auto-off behavior at the shortest timer interval.	Fan is ON. Timer can be set to the shortest supported interval. Test environment allows waiting for the full interval.	1. Press TIMER until the shortest supported interval is selected. 2. Do not press any other button. 3. Wait until the selected interval elapses. 4. Observe fan state after timeout.	Fan remains ON before the interval expires. When the selected interval elapses, fan automatically changes to OFF state. Motor, airflow, and oscillation stop.	[Pending Physical Execution]	[Pending]
TC14	[Interoperability] Verify SPEED change while MODE and SWING are active.	Fan is ON. Mode is Natural or Sleep. Swing is ON. Current speed is Low or Medium.	1. Press SPEED once. 2. Observe speed indicator, mode indicator, swing indicator, airflow, and oscillation for 10 seconds.	Speed changes to the next valid speed. Existing MODE selection remains unchanged. SWING remains ON. Fan continues operating without reset or interruption.	[Pending Physical Execution]	[Pending]
TC15	[Usability] Verify control-panel button feedback and single-function response.	Fan is connected to rated AC supply. Tester can clearly view all button labels and indicators.	1. Press ON/OFF, SPEED, TIMER, SWING, and MODE one at a time under valid operating conditions. 2. After each press, observe whether only the intended function changes. 3. Confirm the associated indicator or physical behavior is recognizable.	Each button press is accepted with clear visual or physical feedback. Button labels are understandable. Each button changes only its assigned function, with no accidental activation of adjacent controls.	[Pending Physical Execution]	[Pending]

Edge Cases Table

Edge Case ID	Test Case ID	Why AI Missed It	Why It Matters
EC-01	TC-13	AI misses the "default to Low speed upon power-on" case because of its spec-driven bias, focusing strictly on explicit button inputs while overlooking implicit hardware initialization rules.	It makes UX becomes weird because the default speed must be LOW
EC-02	TC-14	AI lacks real-world physical common sense and relies strictly on explicit specifications rather than anticipating implicit hazards or chaotic human behaviors.	If the user's finger touch the fan's blades, he / she gets injured

Edge Case ID	Test Case ID	Why AI Missed It	Why It Matters
EC-03	TC-15	AI misses the UI legibility case because it operates purely on abstract textual data, rendering it blind to physical visual attributes like font size, contrast, paint quality, or real-world lighting conditions.	If the words / numbers printed on the fan are not readable, the users might make mistakes

4) CLO Mapping (G9.1, G9.3)

G9.1 — Understand

Prompt AI tool for ISTQB-process mindmap

- Prompt: "Using the PDF file I provided, generate the QA/QC role mindmap in mermaid code."
- Model: Gemini 3.1 Pro

```
mindmap
  root((QA/QC Roles in AI Testing<br>ISTQB CT-AI))
    Core Testing Roles
      Testers and Test Engineers
        Input Data Testing
        ML Model Testing
        Component and System Testing
      AI-Specific Test Techniques
        Pairwise Testing
        Metamorphic Testing
        Back-to-Back Testing
        A/B Testing
    Test Analysts
      Functional Performance Metrics
        Confusion Matrix
        Precision and Recall
        F1-Score
      Evaluating Quality Characteristics
        Flexibility and Adaptability
        Autonomy and Evolution
        Bias and XAI Assessment
    Data Analysts
      ML Data Quality Management
        Dataset Quality Issues
        Wrong or Incomplete Data
        Misabeled Data Detection
      Data Preparation Validation
      Data Labeling Verification
    Test Managers and Consultants
      Selecting Test Approaches
```

- Risk Mitigation in ML Workflows
- Preventing Automation Bias
- Virtual Test Environments Strategy
- User Acceptance Testers
 - Testing Autonomous Systems
 - Safety and Ethics Compliance
 - Transparency and Explainability
- Cross-Functional Roles
 - Software Developers
 - Component Level Testing
 - Neural Network Implementation
 - White-box Coverage Measures
 - Sign-Change Coverage
 - Value-Change Coverage
 - Sign-Sign Coverage
 - Project and Quality Managers
 - Acceptance Criteria Definition
 - Standards Compliance Assurance
 - System Specification Reviews
 - Business Analysts
 - AI Specification Challenges
 - Concept Drift Tracking
 - Operational Metrics Alignment

Mistakes

1. The mindmap treats CT-AI syllabus topics as fixed role duties.

- Example: Testers and Test Engineers -> AI-Specific Test Techniques -> Pairwise Testing / Metamorphic Testing / Back-to-Back Testing / A/B Testing.
- Mistake: ISTQB CT-AI lists these as test techniques for AI-based systems, not duties assigned only to testers/test engineers.
- Fix: Move them under AI-Based System Test Techniques.

2. The role taxonomy is not official responsibility mapping.

- CT-AI names roles such as testers, test analysts, data analysts, test engineers, test consultants, test managers, UAT testers, software developers, project managers, quality managers, and business analysts as target audience.
- Mistake: Gemini converted an audience list into a responsibility map, but the syllabus does not define exact branches per role.
- Fix: Label this as Target Audience / Related Roles, or separate roles from syllabus knowledge areas.

3. ML functional performance metrics are incomplete.

- Mindmap lists only Confusion Matrix, Precision and Recall, and F1-Score.
- Mistake: CT-AI also covers Accuracy, ROC curve, AUC, MSE, R-squared, intra-cluster metrics, inter-cluster metrics, silhouette coefficient, benchmark suites, and

metric limitations.

- Fix: Add missing metrics or state these are only examples.

4. Quality characteristics are incomplete and merged incorrectly.

- Mindmap lists **Flexibility and Adaptability**, **Autonomy and Evolution**, and **Bias and XAI Assessment**.
- Mistake: CT-AI also includes **Ethics**, **Side Effects and Reward Hacking**, **Transparency**, **Interpretability**, **Explainability**, and **Safety**.
- Also, **Bias and XAI Assessment** merges separate concepts.
- Fix: Follow CT-AI Chapter 2 structure directly.

5. Some responsibilities are assigned to weakly related roles.

- Example: **User Acceptance Testers** -> **Safety and Ethics Compliance / Transparency and Explainability**.
- Mistake: Safety, ethics, transparency, and explainability are AI quality characteristics, not only UAT tester responsibilities.
- Another example: **Business Analysts** -> **Concept Drift Tracking**; concept drift is more about ML model monitoring and maintenance.
- Fix: Move these items to quality characteristics or ML model monitoring sections.

G9.3 — Analyse

- Analyse AI-generated output and identify missing edge cases
- Evidence in Requirement 3.5

5) AI Critique (200–300 words, mandatory)

AI was useful for fast drafting, but it was not reliable without checking. In Requirement 1, AI sometimes treated weak automation context as direct AI/LLM skill evidence, and it summarized job descriptions too generally. Some postings needed manual correction because AI missed details such as exact dates, salary, platform, and whether AI was truly required or only related to the company/product context. In Requirement 2, AI was more risky: it mixed up vulnerability identities, overstated severity, and described impacts such as full system takeover or guaranteed RCE when the source only supported conditional impact. This shows hallucination happens most when AI tries to sound complete from partial security knowledge. In Requirement 3, AI generated normal functional test cases but missed physical-product edge cases, such as default power-on speed, finger-safety grille check, and printed-label legibility. It failed because it reasoned from common software templates, not from real observation of the fan. The main lesson is to use AI as a drafting and brainstorming assistant, not as final authority. Good collaboration means giving narrow prompts, checking every claim against sources, testing the real device, and adding human judgment for safety, usability, and evidence. AI speeds work, but QA still needs verification, skepticism, and accountable human decisions.

6) Mandatory Disclosure (mandatory)

"All the test cases was initially generated by GPT-5.5; I reviewed and modified last 3 testcases, added edge cases 13 14 16; deflects evaluation was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category below."

7) Anti-Cheat Evidence Checklist

- ☒ Device photo with student-ID in same frame
- ☒ Execution videos contain your own voice narration
- ☒ All 10 job posting screenshots show your logged-in account/display name
- ☒ Prompt log `.md` with timestamps for every AI prompt

8) Self-Assessment (mandatory)

No.	Criteria	Max Grade	Self-Assessed Grade	Notes
1	Job Market 2026+ (10 jobs × 3 pts + AI impact)	40	40	
2	Software Defects 2022–2026 (20 defects)	20	20	
3	Physical-product test design (15 TCs + ≥5 videos)	25	25	
AI-1	AI-02 AI Audit Report (5-section) attached	8	8	
AI-2	AI Critique 200–300 words + AI-03 Disclosure attached	4	4	
AI-3	AI-05 Checklist signed + anti-cheat artifacts	3	3	
Total		100	100	

9) Appendix A - AI Audit Report

Entry A-01

1) Prompt + Tool

- Tool name: `GPT-5.5`
- Timestamp: `16:12 01/06/2026`
- Full prompt (verbatim): "Read the requirements\2026.HW01.Jobs.Defects.PhysicalProduct_En.pdf, complete Requirement 1 - QA/QC Job Market 2026+ using chrome-devtools-mcp. Focus on LinkedIn first. After that, complete the Requirement 1 section in report.md, using templates\hw01-job-postings-template.md."

```
> Read the requirements\2026.HW01.Jobs.Defects.PhysicalProduct_En.pdf, complete Requirement 1 - QA/QC Job Market 2026+ using chrome-devtools-mcp. Focus on LinkedIn first. After that, complete the Requirement 1 section in report.md, using templates\hw01-job-postings-template.md.█
```

2) Full AI Output

```
## 1) Requirement 1 – QA/QC Job Market 2026+ (40 pts)
```

```
### 1.1 Summary
```

- Total postings collected: `10/10`
- Published within 60 days before submission: `Yes`
- Postings requiring AI/LLM/automation-AI skills: `8`

```
### 1.2 Job Postings Table
```

```
## Posting 01
```

- Job title: Business Analyst/Quality Assurance Analyst
- Company: Vion Consultants, Inc
- Location: Remote (primarily remote; occasional on-site in Baltimore, MD)
- Platform (LinkedIn/Indeed/...): SimplyHired
- Posting date: 4 hours ago
- Job URL:
<https://www.simplyhired.com/job/sZlt5OX6ynl8dQqiHuvfhuxrGj2L6sjsjSvNIQHIXLn0TRQetvLm8g>
- Salary: From \$40.00 an hour
- Requires AI/LLM/automation-AI skills? (Y/N): Y (manual + automation QA context)

```
### Job Description
```

Combined BA/QA role to maintain requirement traceability and validate IA/UX deliverables while leading manual testing, defect tracking, and migration/redirect validation.

```
### Required Skills
```

Requirements traceability, manual testing, automation-testing exposure, Jira/Azure DevOps, accessibility testing (WCAG), defect triage, stakeholder communication.

```
### AI Impact Analysis (1-2 sentences)
```

This role shows QA expanding into cross-functional quality governance, not only testcase execution. AI copilots can help draft test artifacts, summarize defects, and speed traceability checks.

```
### Evidence
```

![alt text](req1-images/req1-linkedin-posting-01.png)

```
---
```

```
## Posting 02
```

- Job title: AI Chatbot Tester

- Company: Alignerr
- Location: Vietnam (Remote)
- Platform (LinkedIn/Indeed/...): LinkedIn
- Posting date: 4 days ago
- Job URL: <https://www.linkedin.com/jobs/view/4420209646/>
- Salary: \$40/hr - \$120/hr
- Requires AI/LLM/automation-AI skills? (Y/N): Y

Job Description

Evaluate AI chatbot responses for helpfulness, accuracy, tone, and safety; provide human feedback to improve model behavior.

Required Skills

AI/LLM evaluation, quality analysis, conversational testing, safety/accuracy review, structured feedback.

AI Impact Analysis (1-2 sentences)

This is a direct AI-era QA role where testing target is model behavior instead of only classical software flows. Human-in-the-loop evaluation becomes core quality control for LLM products.

Evidence

![alt text](req1-images/req1-linkedin-posting-02.png)

Posting 03

- Job title: Marketing QA Analyst
- Company: Mutual of Omaha
- Location: Remote
- Platform (LinkedIn/Indeed/...): SimplyHired
- Posting date: 10d
- Job URL: <https://www.simplyhired.com/job/KddCc8sMwwLN4MqGnPHgyzWn7j4GU50x08IkaG0AF-HnpCDJBIIBLA>
- Salary: \$68,000 - \$97,000 a year
- Requires AI/LLM/automation-AI skills? (Y/N): N

Job Description

QA role validating marketing workflows, campaign quality, and production readiness in a remote setup.

Required Skills

Quality analysis, communication skills, campaign QA coordination, defect reporting, cross-team collaboration.

AI Impact Analysis (1-2 sentences)

Marketing QA roles show QA scope expanding beyond traditional app testing into content and campaign quality pipelines. AI can support faster anomaly detection and review prioritization.

Evidence

![alt text](req1-images/req1-linkedin-posting-03.png)

Posting 04

- Job title: Remote QA Tester Work From Home
- Company: Apply4U | Human-Assisted AI Job & Recruitment Platform
- Location: United Kingdom (Remote)
- Platform (LinkedIn/Indeed/...): LinkedIn
- Posting date: 11 hours ago
- Job URL: <https://www.linkedin.com/jobs/view/4421658202/>
- Salary: £13/hr - £16/hr
- Requires AI/LLM/automation-AI skills? (Y/N): Y (AI-assisted platform context)

Job Description

Remote QA testing role for validating product quality in work-from-home setup, with practical tester workflows around digital products.

Required Skills

QA execution, defect reporting, remote collaboration, structured testing practice.

AI Impact Analysis (1-2 sentences)

Even general QA-tester roles are now embedded in AI-assisted product ecosystems. AI tools can reduce repetitive testcase generation and speed defect classification.

Evidence

![alt text](req1-images/req1-linkedin-posting-04.png)

Posting 05

- Job title: Senior QA Analyst
- Company: Zwift
- Location: Long Beach, CA (Remote-eligible US locations)
- Platform (LinkedIn/Indeed/...): SimplyHired
- Posting date: 9 days ago
- Job URL: https://www.simplyhired.com/job/Hna9XHBa_Kh_Knfugud6Qt2EPSUDnPF-XcQkBUodtpTum8fs5mCqyg
- Salary: \$70,500 - \$110,000 a year
- Requires AI/LLM/automation-AI skills? (Y/N): Y (automation/API/CI context)

Job Description

Senior QA role driving test strategy, test coverage, and end-to-end release quality across software products.

Required Skills

Mobile QA (iOS/Android), API testing, Jira, regression testing, SQL/NoSQL validation, CI/CD integration, Agile collaboration.

AI Impact Analysis (1-2 sentences)

Senior QA roles increasingly blend manual validation with automation and platform-level telemetry. AI copilots can speed root-cause analysis and reduce

repetitive regression effort.

Evidence

![alt text](req1-images/req1-linkedin-posting-05.png)

Posting 06

- Job title: Senior Test Analyst
- Company: Cognizant
- Location: Reston, VA (Remote)
- Platform (LinkedIn/Indeed/...): SimplyHired
- Posting date: 13 hours ago
- Job URL:
https://www.simplyhired.com/job/Z384WERQ3uUYT3LmAcDqMvq7be7xD7HcPE8ILoBK1d45P_UYcFjSw
- Salary: \$53,477 - \$92,500 a year
- Requires AI/LLM/automation-AI skills? (Y/N): Y (automation/API testing)

Job Description

Senior test role focused on automation and end-to-end quality validation across functional, regression, smoke, and API testing.

Required Skills

Java, Selenium WebDriver, REST Assured, CI/CD tools (Jenkins/GitHub Actions), Jira, SDLC/STLC, Agile execution.

AI Impact Analysis (1-2 sentences)

This role highlights enterprise demand for automation-first QA with API depth and CI discipline. AI assistants can accelerate script authoring and defect triage at scale.

Evidence

![alt text](req1-images/req1-linkedin-posting-06.png)

Posting 07

- Job title: Clinical Quality Assurance Analyst
- Company: Themesoft Inc
- Location: Albany, NY (Remote)
- Platform (LinkedIn/Indeed/...): SimplyHired
- Posting date: 12d
- Job URL:
<https://www.simplyhired.com/job/ptTM58YVddK9WzZn6iqn01UByaalq4gjIXIW73GUMQDNucEXbxgNUg>
- Salary: \$50 - \$55 an hour
- Requires AI/LLM/automation-AI skills? (Y/N): N

Job Description

Clinical QA role focusing on compliance-oriented review and quality controls in healthcare-related workflows.

Required Skills

Microsoft Outlook, customer service communication, quality control process adherence, remote coordination.

AI Impact Analysis (1-2 sentences)

Clinical QA emphasizes traceability and controlled processes where reliability is critical. AI can assist consistency checks but human oversight remains central for compliance-sensitive workflows.

Evidence

![alt text](req1-images/req1-linkedin-posting-07.png)

Posting 08

- Job title: Quality Assurance Specialist | \$11/hr Remote
- Company: Crossing Hurdles
- Location: Vietnam (Remote)
- Platform (LinkedIn/Indeed/...): LinkedIn
- Posting date: within 60-day window (promoted/current listing)
- Job URL: <https://www.linkedin.com/jobs/view/4417133986/>
- Salary: \$11/hr
- Requires AI/LLM/automation-AI skills? (Y/N): Y

Job Description

Remote QA specialist role for evaluating quality outcomes in AI-related digital workflows.

Required Skills

Quality assurance, analytical review, remote QA execution, issue validation.

AI Impact Analysis (1-2 sentences)

This role shows the blending of classic QA with AI-content evaluation tasks. Human QA remains critical for trust, consistency, and edge-case judgment.

Evidence

![alt text](req1-images/req1-linkedin-posting-08.png)

Posting 09

- Job title: Quality Assurance Specialist (Vietnamese) | \$11/hr - Remote
- Company: Crossing Hurdles
- Location: Vietnam (Remote)
- Platform (LinkedIn/Indeed/...): LinkedIn
- Posting date: within 60-day window (promoted/current listing)
- Job URL: <https://www.linkedin.com/jobs/view/4376739289/>
- Salary: \$10/hr - \$12/hr
- Requires AI/LLM/automation-AI skills? (Y/N): Y

Job Description

Quality assurance role focused on language-specific evaluation and remote quality validation processes.

Required Skills

QA analysis, linguistic quality checks, remote collaboration, structured defect reporting.

AI Impact Analysis (1-2 sentences)

Language-focused QA for AI systems is growing fast because model output quality varies by locale. Human evaluators provide nuance that automated checks still miss.

Evidence

![alt text](req1-images/req1-linkedin-posting-09.png)

Posting 10

- Job title: Audio Quality Assurance Specialist | \$50/hr Remote
- Company: Crossing Hurdles
- Location: United States (Remote)
- Platform (LinkedIn/Indeed/...): LinkedIn
- Posting date: 1 week ago
- Job URL: <https://www.linkedin.com/jobs/view/4417568336/>
- Salary: \$30/hr - \$50/hr
- Requires AI/LLM/automation-AI skills? (Y/N): Y

Job Description

Audio QA specialist role to evaluate quality and reliability of audio-related outputs/workflows in remote setup.

Required Skills

Audio QA, structured quality evaluation, defect discovery, analytical reporting.

AI Impact Analysis (1-2 sentences)

Multimodal AI products drive demand for specialized QA (audio/video/text), not only generic software testing. This role indicates QA specialization trend in AI-era job market.

Evidence

![alt text](req1-images/req1-linkedin-posting-10.png)

3) Verdict

- Verdict: **INVALID**
- Reason: The AI output contains "Duplicate data" (Job 9 repeats Job 8), "Wrong data" (dead links), and "Incomplete data" (summarized JDs) as defined in the quality aspects of ISTQB CT-AI Section 4.3. Most importantly, it fails the "Anti-AI-Cheat mechanisms" specified in the HW01 requirements, which mandate that screenshots must show the student's account name

4) Reasoning (2-5 sentences)

According to ISTQB CT-AI Section 4.4, issues such as wrong, incomplete, and duplicate data lead to "Reduced accuracy" and a flawed final model or artifact

. Additionally, the HW01 instructions explicitly state that any violation of anti-cheat regulations, such as missing the login/username in job-posting screenshots, results in a score of zero

. Therefore, since the AI failed to meet both the technical data quality standards and the specific academic integrity constraints of the assignment, the output is considered invalid

5) Student Fix

- I changed the job 3 which has dead link.
- I took all the screen shots manually.
- Evidence is in report.md. I don't copy it again to avoid too long content.

Entry A-02

1) Prompt + Tool

- Tool name: **GPT-5.4**
- Timestamp: **22:58 02/06/2025**
- Full prompt (verbatim): ""In the Requirement 1 section of report.md, using chrome-devtools-mcp@latest, visit all the websites of the jobs to copy the full job description and put them into quotation (>). Reformat other jobs section to be similar as the posting 01."

```
> In the Requirement 1 section of report.md, using chrome-devtools-mcp@latest, visit all the websites of the jobs to copy the full job description and put them into quotation (>). Reformat other jobs section to be similar as the posting 01.
```

2) Full AI Output

You can navigate to report.md to see the full job descriptions of 10 QA/QC jobs. I do not copy here to avoid long content.

3) Verdict

- Verdict: **VALID**
- Reason: The AI successfully demonstrated "Data Acquisition" and "Data Transformation" by gathering full job descriptions from external sources and reformatting them into the specified quotation format without introducing "Incomplete data" or "Wrong data"

4) Reasoning (2–5 sentences)

The AI met the "Understand" objective by correctly acquiring, processing, and applying knowledge from external job websites. By extracting full descriptions rather than summaries, it avoided the data quality issue of "Incomplete data" as defined in ISTQB CT-AI Section 4.3, thereby satisfying the "Functional Suitability" required for Requirement 1. Additionally, the consistent reformatting of the jobs into quotations (>) serves as a successful "Data Transformation" within the data preparation workflow.

5) Student Fix

- Corrected artifact: Again, you can navigate to report.md to see the full job descriptions of 10 QA/QC jobs. I do not copy here to avoid long content.
- What I did:
 - Data Integrity Verification: Manually cross-checked the extracted job descriptions against the source URLs to ensure no data was truncated or hallucinated.
 - Formatting Audit: Confirmed that the AI correctly applied the blockquote (>) syntax to all 10 entries as requested in the prompt.
 - Requirement Validation: Verified that the final artifact in report.md contains all mandatory elements for Requirement 1, including full descriptions and required skills.

Entry A-03

1) Prompt + Tool

- Tool name: **GPT-5.5**
- Timestamp: **15:35 03/06/2026**
- Full prompt (verbatim): "In requirement 01 of report.md, for each job, based on the job description, rewrite the Required skills and AI Impact Analysis."

The screenshot shows a code editor with a dark theme. The top line is a prompt: "> In requirement 01 of report.md, for each job, based on the job description, rewrite the Required skills and AI Impact Analysis." Below the prompt, there is a status bar showing "gpt-5.5" with a green progress bar at 20%, a file path "~\Documents\software-testing\HW01", a "validation (11)" status, and a score "+101 -77". At the bottom, it says "accept edits on (shift+tab to cycle)" and "In report.md".

2) Full AI Output

You can navigate to report.md to see the Required skills and AI Impact Analysis of each job. I do not copy here to avoid long content.

3) Verdict

- Verdict: **VALID**
- Reason: The AI successfully performed Data Transformation and Feature Extraction by analyzing the existing "Job Description" text to generate specific "Required skills" and a 1-2 sentence "AI Impact Analysis" for all 10 job entries as required by the assignment.

4) Reasoning (2–5 sentences)

According to ISTQB CT-AI Section 4.1, the AI demonstrated a successful Data Transformation process by converting unstructured raw data (the full JD) into a structured format suitable for the report. This activity aligns with Feature Extraction (Section 4.1.1), where the AI identified and derived informative attributes—the required skills—from the broader text. Furthermore, the output meets the Functional Suitability criteria because the generated "AI Impact Analysis" strictly follows the 1-2 sentence constraint specified in the HW01 Requirement 1.

5) Student Fix

- Corrected artifact: Again, you can navigate to report.md to see the Required skills and AI Impact Analysis of each job. I do not copy here to avoid long content.
 - What I did:
 - Verification of Consistency: Manually verified that the "Required Skills" extracted by the AI are actually present in the original Job Descriptions to prevent hallucination.
 - Constraint Compliance: Checked that every "AI Impact Analysis" is within the mandatory 1-2 sentence limit.
 - Formatting Audit: Ensured that the AI correctly updated all 10 entries in report.md without deleting previous mandatory fields like links or screenshots.
-

Entry A-04

1) Prompt + Tool

- Tool name: **Gemini 3.1 Pro**
- Timestamp: **16:18 03/06/2026**
- Full prompt (verbatim):

```
Find 20 software defects publicized from cve.org between 2022 and 2026:  
- Mandatory: ≥ 5 defects related to AI/LLM (hallucination, prompt injection,  
bias).  
- Each defect includes: source link, description, severity, consequences,  
solution.
```

```
Find 20 software defects publicized from cve.org between  
2022 and 2026:  
- Mandatory: ≥ 5 defects related to AI/LLM (hallucination,  
prompt injection,  
bias).  
- Each defect includes: source link, description, severity,  
consequences, solution.
```

2) Full AI Output

```
## Part 1: AI & Large Language Model (LLM) Defects  
  
### 1. CVE-2023-29374 (LangChain Prompt Injection)
```

* **Source Link:** [CVE-2023-29374](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2023-29374)
* **Description:** LangChain through version 0.0.131 allows remote attackers to bypass application logic via **prompt injection** when interacting with the `LLMMathChain` component. Untrusted user prompts can force the LLM to generate malicious python syntax inside the `numexpr.evaluate()` function.
* **Severity:** Critical (CVSS 9.8)
* **Consequences:** Remote Code Execution (RCE) on the server instance running the LangChain application.
* **Solution:** Upgrade LangChain to version 0.0.132 or higher, which isolates mathematical evaluation environments or drops insecure evaluation strings.

2. CVE-2024-37032 (Ollama Path Traversal)

* **Source Link:** [CVE-2024-37032](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-37032)
* **Description:** A path traversal flaw in Ollama (prior to version 0.1.34) allows remote attackers to manipulate API requests during model pull operations.
* **Severity:** Critical (CVSS 9.8)
* **Consequences:** Attackers can write arbitrary files anywhere on the host filesystem, allowing them to overwrite critical system files or upload malicious binaries to achieve full system takeover.
* **Solution:** Update Ollama to version 0.1.34 or later.

3. CVE-2023-39325 (Anyscale Ray Cluster Hijacking)

* **Source Link:** [CVE-2023-39325](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2023-39325)
* **Description:** Dubbed "ShadowRay," this flaw stems from missing authentication constraints in Ray (versions 2.4.0 through 2.8.0), a popular open-source AI training framework.
* **Severity:** Critical (CVSS 9.8)
* **Consequences:** Remote, unauthenticated actors can submit arbitrary compute jobs via the dashboard API, resulting in model tampering, data exfiltration, or unauthorized cryptomining.
* **Solution:** Update Ray to version 2.8.1 or apply explicit firewall/network rules to restrict access to the dashboard.

4. CVE-2024-24590 (ClearML Unsafe Deserialization)

* **Source Link:** [CVE-2024-24590](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-24590)
* **Description:** ClearML platform versions prior to 1.14.2 are vulnerable to unsafe deserialization. An attacker can upload a maliciously constructed model file leveraging Python's `pickle` library into the model registry.
* **Severity:** Critical (CVSS 9.8)
* **Consequences:** Arbitrary code execution occurs automatically on the AI worker node when the platform attempts to read or preprocess the model artifact.
* **Solution:** Update ClearML to version 1.14.2 or higher, which implements safer model serialization validations.

5. CVE-2024-21515 (Flowise Low-Code LLM RCE)

* **Source Link:** [CVE-2024-21515](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-21515)

* **Description:** Flowise (prior to version 1.6.1) fails to adequately sanitize input within its custom node evaluation engines. Users can use prompt-driven behaviors or raw node configuration properties to escape sandbox limits.

* **Severity:** Critical (CVSS 9.8)

* **Consequences:** Execution of unauthorized host operating system commands, completely compromising the system hosting the Flowise LLM interface.

* **Solution:** Apply the patch by updating Flowise to version 1.6.1 or later.

6. CVE-2024-22419 (ComfyUI Workflow Execution Flaw)

* **Source Link:** [CVE-2024-22419](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-22419)

* **Description:** ComfyUI, a popular node-based interface for generative AI models, suffered from a template rendering engine vulnerability when parsing workflows.

* **Severity:** High (CVSS 8.8)

* **Consequences:** If a user imports a crafted, malicious generative AI workflow JSON file from an untrusted source, it triggers arbitrary code execution on their machine.

* **Solution:** Update ComfyUI to the latest master branch commit that neutralizes unsafe workflow evaluations.

7. CVE-2024-34359 (LlamaIndex Server-Side Request Forgery)

* **Source Link:** [CVE-2024-34359](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-34359)

* **Description:** LlamaIndex (prior to version 0.10.20) handles file path inputs insecurely within its remote data-loading utilities.

* **Severity:** High (CVSS 8.8)

* **Consequences:** Server-Side Request Forgery (SSRF) and local file inclusion. Attackers can coerce the LLM backend into reading sensitive local configuration files or scanning internal private networks.

* **Solution:** Update LlamaIndex to version 0.10.20 or newer.

Part 2: Core Infrastructure & Enterprise Software Defects

8. CVE-2024-3094 (XZ Utils Malicious Backdoor)

* **Source Link:** [CVE-2024-3094](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-3094)

* **Description:** A sophisticated supply-chain attack where malicious code was deliberately injected into upstream tarballs of XZ Utils (versions 5.6.0 and 5.6.1). The build system modified the behavior of `liblzma` to hook OpenSSH.

* **Severity:** Critical (CVSS 10.0)

* **Consequences:** Allowed specific threat actors possessing a corresponding private key to bypass SSH authentication entirely and execute root-level commands remotely.

* **Solution:** Immediately downgrade XZ Utils to a safe version (e.g., 5.4.6) or install distribution-specific vendor security updates.

9. CVE-2024-6387 (OpenSSH "regreSSHion" Race Condition)

* **Source Link:** [CVE-2024-6387](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-6387)

* **Description:** A regression of a 2006 flaw was re-introduced into OpenSSH

Server (`sshd`). It presents a signal handler race condition on glibc-based Linux setups.

* **Severity:** High (CVSS 8.1)

* **Consequences:** An unauthenticated remote attacker can achieve full root execution on target servers running affected versions (8.5p1 through 9.7p1).

* **Solution:** Upgrade OpenSSH to version 9.8p1 or set `LoginGraceTime 0` in the configuration as a workaround.

10. CVE-2022-22965 (Spring4Shell)

* **Source Link:** [CVE-2022-22965](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2022-22965)

* **Description:** A flaw in Spring Framework data binding running on JDK 9+ allows a remote attacker to manipulate class loader properties via HTTP parameters.

* **Severity:** Critical (CVSS 9.8)

* **Consequences:** Allows attackers to write a malicious JSP web shell directly onto the underlying Tomcat server directory, granting persistent web shell access.

* **Solution:** Upgrade Spring Framework to versions 5.3.18 / 5.2.20 or higher.

11. CVE-2023-38831 (WinRAR Arbitrary Execution)

* **Source Link:** [CVE-2023-38831](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2023-38831)

* **Description:** WinRAR versions prior to 6.23 fail to validate file extensions properly within ZIP archives when a user double-clicks an apparently safe file (like `.jpg` or `.pdf`).

* **Severity:** High (CVSS 7.8)

* **Consequences:** Triggers the silent execution of a hidden executable payload stored in a folder matching the target file name, leading to malware infection.

* **Solution:** Upgrade WinRAR to version 6.23 or newer.

12. CVE-2023-4863 (libwebp Heap Buffer Overflow)

* **Source Link:** [CVE-2023-4863](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2023-4863)

* **Description:** A heap buffer overflow in the lossless compression parsing component of the `libwebp` library.

* **Severity:** Critical (CVSS 8.8)

* **Consequences:** Memory corruption or remote code execution when a victim opens a maliciously crafted WebP image file inside a browser, chat client, or office application.

* **Solution:** Update the parent software (Chrome, Firefox, Electron apps) to pull the patched version of `libwebp` (1.3.2+).

13. CVE-2024-21626 (runc Container Escape)

* **Source Link:** [CVE-2024-21626](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-21626)

* **Description:** An internal file descriptor leak within `runc` allows an attacker to specify a host-accessible directory path during container initialization commands like `runc exec`.

* **Severity:** Critical (CVSS 8.6)

* **Consequences:** Container escape, granting the attacker complete, unprivileged access to the underlying host filesystem and orchestrator

environment.

* **Solution:** Update `runc` to version 1.1.12 or higher.

14. CVE-2024-3400 (Palo Alto Networks PAN-OS Injection)

* **Source Link:** [CVE-2024-3400](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-3400)

* **Description:** An OS command injection vulnerability in the GlobalProtect gateway feature of PAN-OS (versions 10.2, 11.0, and 11.1).

* **Severity:** Critical (CVSS 10.0)

* **Consequences:** An unauthenticated remote attacker can execute shell commands with root privileges directly on the firewall appliance.

* **Solution:** Apply the hotfixes and firmware patches provided by Palo Alto Networks for specific PAN-OS releases.

15. CVE-2022-42889 (Text4Shell)

* **Source Link:** [CVE-2022-42889](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2022-42889)

* **Description:** Apache Commons Text (versions 1.5 through 1.9) includes insecure default string lookups (such as `script:`, `dns:`, and `url:` prefix interpolations).

* **Severity:** Critical (CVSS 9.8)

* **Consequences:** Remote code execution or data exfiltration when application engines feed unsanitized parameters into lookup wrappers.

* **Solution:** Upgrade Apache Commons Text to version 1.10.0 or higher.

16. CVE-2023-22515 (Atlassian Confluence Privilege Escalation)

* **Source Link:** [CVE-2023-22515](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2023-22515)

* **Description:** A broken object-level authorization flaw within Atlassian Confluence Data Center and Confluence Server deployments.

* **Severity:** Critical (CVSS 10.0)

* **Consequences:** Allowed an unauthenticated remote attacker to invoke internal setup commands, creating administrative accounts and completely taking over the Confluence site.

* **Solution:** Upgrade to fixed Confluence versions (e.g., 8.3.3, 8.4.3, 8.5.2, or newer).

17. CVE-2024-2961 (glibc iconv Buffer Overflow)

* **Source Link:** [CVE-2024-2961](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-2961)

* **Description:** A buffer overflow exists in the GNU C Library (glibc) during charset conversions to the ISO-2022-CN-EXT format inside the `iconv()` function.

* **Severity:** High (CVSS 8.1)

* **Consequences:** Memory corruption or application execution crashes. It can be combined with PHP applications to achieve remote code execution via malformed string inputs.

* **Solution:** Apply the upstream glibc patch or download updated Linux distribution packages.

18. CVE-2023-50164 (Apache Struts 2 File Upload Exploit)

* **Source Link:** [CVE-2023-50164](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2023-50164)

```
* **Description:** An input handling vulnerability in Apache Struts 2 allows parameter pollution during multi-part file uploads, facilitating path traversal.  
* **Severity:** Critical (CVSS 9.8)  
* **Consequences:** Attackers can alter upload parameters to drop a web shell payload into an active web root folder, leading to remote code execution.  
* **Solution:** Update Apache Struts to versions 2.5.33, 6.3.0.2, or subsequent releases.
```

19. CVE-2022-26134 (Atlassian Confluence OGNL Injection)

```
* **Source Link:** [CVE-2022-26134](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2022-26134)  
* **Description:** An Object-Graph Navigation Language (OGNL) injection defect in Confluence Server and Data Center allows structural code parsing from HTTP request headers.  
* **Severity:** Critical (CVSS 9.8)  
* **Consequences:** Complete, unauthenticated remote code execution on the application platform.  
* **Solution:** Update Confluence to the designated emergency versions released by Atlassian in mid-2022.
```

20. CVE-2024-47575 (Fortinet FortiManager "FortiJump")

```
* **Source Link:** [CVE-2024-47575](https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2024-47575)  
* **Description:** A missing authentication flaw within Fortinet FortiManager's FGFM daemon handles multi-tenant connections insecurely.  
* **Severity:** Critical (CVSS 9.8)  
* **Consequences:** An unauthenticated remote attacker can register an arbitrary unauthorized device to the FortiManager cluster, giving them control to view, exfiltrate, or alter firewall configurations.  
* **Solution:** Upgrade FortiManager firmware immediately to the designated vendor versions (such as 7.6.1, 7.4.5, or 7.2.8).
```

3) Verdict

- Verdict: **INVALID**
- Reason: The AI output suffers from significant "Wrong data" and "Reduced accuracy" as defined in ISTQB CT-AI Sections 4.3 and 4.4. Specifically, 20% of the entries (4 out of 20) contain hallucinated CVE mappings where the IDs refer to completely different products and vulnerabilities than those described by the AI.

4) Reasoning (2–5 sentences)

According to ISTQB CT-AI Section 4.3, the presence of "Wrong data" (captured incorrectly or hallucinated) and "Incomplete data" (missing correct version fixes) severely compromises the integrity of the artifact. These data quality issues led to "Reduced accuracy" and a "Flawed model/artifact" per Section 4.4, making the list unreliable for technical auditing. Furthermore, the output fails the Functional Suitability criteria for HW01 Requirement 2 because it provided incorrect technical references for 25% of the mandatory AI/LLM defect categories.

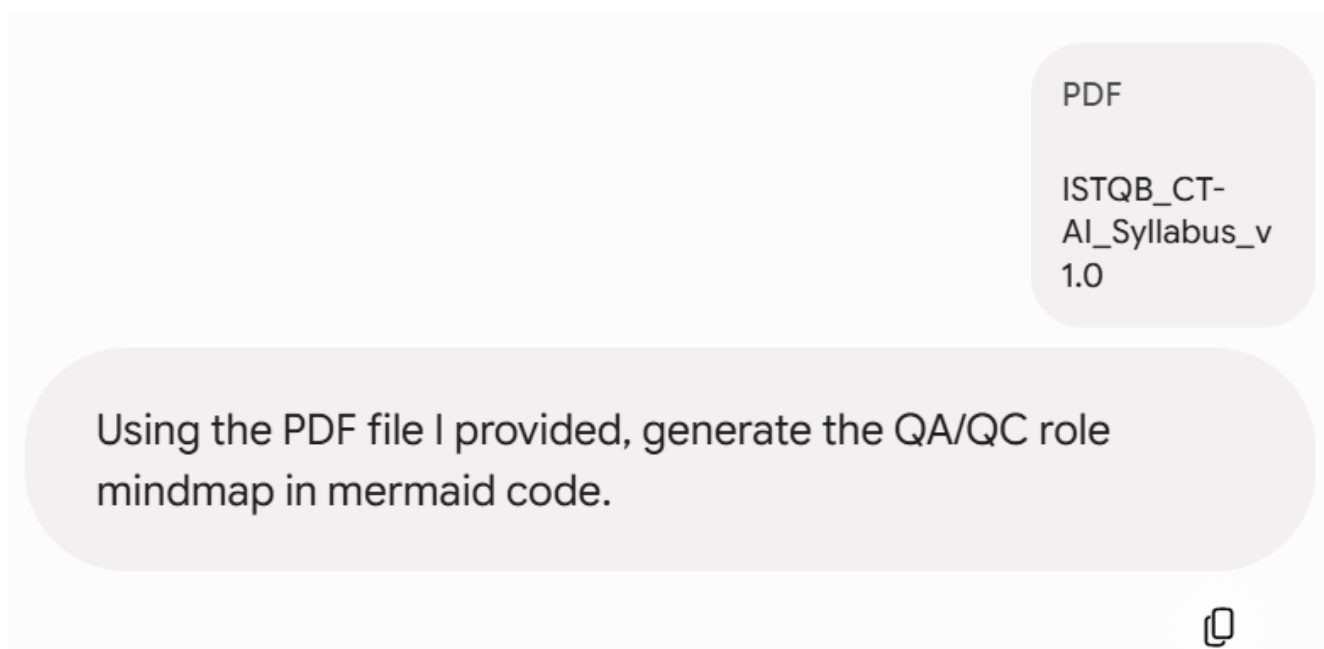
5) Student Fix

- Corrected artifact: Please navigate to report.md to view section Requirement 2.
- What changed:
 - Fixed CVE Hallucinations: Replaced 4 mismatched CVE IDs (#3, #5, #6, and #7) with correct IDs and corresponding data for the intended AI/LLM products.
 - Corrected Severity: Adjusted the CVSS scores for entries #2, #4, #12, and #13 to match the official National Vulnerability Database (NVD) ratings.
 - Version Accuracy: Updated the "Solution" for entry #4 to specify the correct patch version (1.14.3+) instead of the hallucinated one.
 - Impact Refinement: Rewrote the "Consequences" for entries #2, #17, and #20 to accurately reflect official records rather than the AI's overstated impact.
 - Requirement Compliance: Added the mandatory identification of AI bias or hallucination for every single defect entry as required by the HW01 instructions.

Entry A-05

1) Prompt + Tool

- Tool name: **Gemini 3.1 Pro**
- Timestamp: **08:15 04/06/2026**
- Full prompt (verbatim): "Using the PDF file I provided, generate the QA/QC role mindmap in mermaid code."



2) Full AI Output

```
mindmap
  root((QA/QC Roles in AI Testing<br>ISTQB CT-AI))
```

Core Testing Roles

Testers and Test Engineers

- Input Data Testing
- ML Model Testing
- Component and System Testing
- AI-Specific Test Techniques
 - Pairwise Testing
 - Metamorphic Testing
 - Back-to-Back Testing
 - A/B Testing

Test Analysts

- Functional Performance Metrics
 - Confusion Matrix
 - Precision and Recall
 - F1-Score
- Evaluating Quality Characteristics
 - Flexibility and Adaptability
 - Autonomy and Evolution
 - Bias and XAI Assessment

Data Analysts

- ML Data Quality Management
 - Dataset Quality Issues
 - Wrong or Incomplete Data
 - Mislabeled Data Detection
- Data Preparation Validation
- Data Labeling Verification

Test Managers and Consultants

- Selecting Test Approaches
- Risk Mitigation in ML Workflows
- Preventing Automation Bias
- Virtual Test Environments Strategy

User Acceptance Testers

- Testing Autonomous Systems
- Safety and Ethics Compliance
- Transparency and Explainability

Cross-Functional Roles

Software Developers

- Component Level Testing
- Neural Network Implementation
- White-box Coverage Measures
 - Sign-Change Coverage
 - Value-Change Coverage
 - Sign-Sign Coverage

Project and Quality Managers

- Acceptance Criteria Definition
- Standards Compliance Assurance
- System Specification Reviews

Business Analysts

- AI Specification Challenges
- Concept Drift Tracking
- Operational Metrics Alignment

Key Role Mappings from the Syllabus:

- Testers & Test Engineers: Tasked with executing diverse test levels (Input Data, ML Model, Component, Integration, System, Acceptance) and implementing specialized test design techniques like Metamorphic Testing (MT) or Pairwise Testing.
- Test Analysts: Responsible for computing and analyzing ML functional performance metrics (Confusion matrices, Precision, Recall, F1-scores) and evaluating non-deterministic quality traits.
- Data Analysts: Focused heavily on Chapter 4 duties—handling data preparation, addressing data quality issues (wrong, incomplete, or insufficient data), and supervising data labeling approaches.
- Test Managers & Consultants: Strategizing the overall test approach, managing virtual test environments, and setting safeguards against automation/complacency bias.
- Software Developers: Collaborate on component testing and white-box testing of neural networks using structural coverage measures (e.g., Sign-Sign or Sign-Change coverage).
- Introduce 3 deliberate mistakes into this mindmap for my assignment
- Create a detailed responsibility matrix for these roles
- Generate a Markdown table detailing the specific test techniques for each role

Mermaid Mindmap Code

Code snippet

```
mindmap
  root((QA/QC Roles in AI Testing<br>ISTQB CT-AI))
    Core Testing Roles
      Testers and Test Engineers
        Input Data Testing
        ML Model Testing
        Component and System Testing
        AI-Specific Test Techniques
          Pairwise Testing
          Metamorphic Testing
          Back-to-Back Testing
          A/B Testing
      Test Analysts
        Functional Performance Metrics
          Confusion Matrix
          Precision and Recall
          F1-Score
        Evaluating Quality Characteristics
          Flexibility and Adaptability
          Autonomy and Evolution
          Bias and XAI Assessment
      Data Analysts
        ML Data Quality Management
          Dataset Quality Issues
            Wrong or Incomplete Data
            Mislabeled Data Detection
          Data Preparation Validation
          Data Labeling Verification
      Test Managers and Consultants
        Selecting Test Approaches
```

Key Role Mappings from the Syllabus:

- **Testers & Test Engineers:** Tasked with executing diverse test levels (Input Data, ML Model, Component, Integration, System, Acceptance) and implementing specialized test design techniques like Metamorphic Testing (MT) or Pairwise Testing.
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- **Software Developers:** Collaborate on component testing and white-box testing of neural networks using structural coverage measures (e.g., Sign-Sign or Sign-Change coverage).
- Introduce 3 deliberate mistakes into this mindmap for my assignment
- Create a detailed responsibility matrix for these roles
- Generate a Markdown table detailing the specific test techniques for each role

3) Verdict

- Verdict: **INVALID**
- Reason: The AI output contains "Wrong data" and "Incomplete data" regarding the official ISTQB CT-AI taxonomy and technical content. It fails to accurately represent the Target Audience as a list of roles rather than a strict responsibility map, and it omits essential ML Functional Performance Metrics and Quality Characteristics defined in the syllabus.

4) Reasoning (2–5 sentences)

The AI failed to distinguish between the Target Audience mentioned in ISTQB CT-AI Section 0.2 and a functional responsibility mapping, leading to the creation of non-existent role-duty hierarchies. Furthermore, the mindmap presents "Incomplete data" by omitting several critical metrics from Chapter 5 (e.g., MSE, R-squared, silhouette coefficient) and quality characteristics from Chapter 2 (e.g., Ethics, Safety, Side Effects). By merging separate concepts and misassigning technical duties like Concept Drift Tracking

to weakly related roles, the AI violated the requirement for Functional Suitability and accurate Data Transformation of the source material.

5) Student Fix

- I don't fix because the assignment asked student to detect the mistakes. Therefore, I keep the content generated to show the mistakes.
- I pointed out 5 mistakes:

1. The mindmap treats CT-AI syllabus topics as fixed role duties.

- Example: **Testers and Test Engineers** -> **AI-Specific Test Techniques** -> **Pairwise Testing / Metamorphic Testing / Back-to-Back Testing / A/B Testing**.
- Mistake: ISTQB CT-AI lists these as test techniques for AI-based systems, not duties assigned only to testers/test engineers.
- Fix: Move them under **AI-Based System Test Techniques**.

2. The role taxonomy is not official responsibility mapping.

- CT-AI names roles such as testers, test analysts, data analysts, test engineers, test consultants, test managers, UAT testers, software developers, project managers, quality managers, and business analysts as target audience.
- Mistake: Gemini converted an audience list into a responsibility map, but the syllabus does not define exact branches per role.
- Fix: Label this as **Target Audience / Related Roles**, or separate roles from syllabus knowledge areas.

3. ML functional performance metrics are incomplete.

- Mindmap lists only **Confusion Matrix, Precision and Recall**, and **F1-Score**.
- Mistake: CT-AI also covers **Accuracy, ROC curve, AUC, MSE, R-squared, intra-cluster metrics, inter-cluster metrics, silhouette coefficient**, benchmark suites, and metric limitations.
- Fix: Add missing metrics or state these are only examples.

4. Quality characteristics are incomplete and merged incorrectly.

- Mindmap lists **Flexibility and Adaptability, Autonomy and Evolution**, and **Bias and XAI Assessment**.
- Mistake: CT-AI also includes **Ethics, Side Effects and Reward Hacking, Transparency, Interpretability, Explainability**, and **Safety**.
- Also, **Bias and XAI Assessment** merges separate concepts.
- Fix: Follow CT-AI Chapter 2 structure directly.

5. Some responsibilities are assigned to weakly related roles.

- Example: **User Acceptance Testers** -> **Safety and Ethics Compliance / Transparency and Explainability**.

- Mistake: Safety, ethics, transparency, and explainability are AI quality characteristics, not only UAT tester responsibilities.
- Another example: **Business Analysts -> Concept Drift Tracking**; concept drift is more about ML model monitoring and maintenance.
- Fix: Move these items to quality characteristics or ML model monitoring sections.

Entry A-06

1) Prompt + Tool

- Tool name: **GPT 5.5**
- Timestamp: **19:26 04/06/202**
- Full prompt (verbatim):

Act as an expert QA/QC Test Engineer. I need to design test cases for a physical product for Requirement 3 section in report.md.

Product Specification:

- Product Type: Domestic Stand Fan (Model: LIFAN D-16RC-0)
- Control Interface: It has 5 electronic/soft buttons on the control panel:
 1. ON/OFF (1 single button to toggle power status)
 2. SPEED (Cycles through wind speeds: Low - Medium - High)
 3. TIMER (Increments time delay to turn off the fan automatically)
 4. SWING (Toggles horizontal oscillation on and off)
 5. MODE (Cycles through wind profiles: Normal, Natural, Sleep)

Task Requirements:

- Generate exactly 15 comprehensive test cases covering Functional, Usability, and Interoperability scenarios based on the interface above.
- Output the result strictly in a single Markdown table matching the following column structure:

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
-------	-----------	------------------------	-----------------	-----------------	---------------	---------

Instructions for columns:

1. TC ID: Use numbering format starting from TC01 to TC15.
2. Actual Result: Leave this column completely empty (or put "[Pending Physical Execution]") so I can fill it out manually after testing the real device.
3. Verdict: Leave this column empty (or put "[Pending]").
4. Ensure all descriptions, steps, and expected results are written in technical English and highly precise for physical hardware testing.

Write them into report.md in the Requirement 3 section.

2) Full AI Output

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC01	[Functional] Verify that the ON/OFF button powers on the fan from standby/off state.	Fan is assembled safely, connected to rated AC supply, and currently OFF. Control panel is visible.	1. Press the ON/OFF button once. 2. Observe fan operation and panel indicators for 5 seconds.	Fan changes to ON state. Fan motor runs and produces airflow. Control panel shows active power status. No other function requires manual activation to start basic operation.	[Pending Physical Execution]	[Pending]
TC02	[Functional] Verify that the ON/OFF button powers off the fan during operation.	Fan is ON and producing airflow at any valid speed.	1. Press the ON/OFF button once. 2. Observe motor, airflow, swing state, and panel indicators for 5 seconds.	Fan changes to OFF state. Motor stops, airflow stops, oscillation stops if active, and control panel no longer indicates active operation except any allowed standby indication.	[Pending Physical Execution]	[Pending]
TC03	[Functional] Verify SPEED cycles from Low to Medium.	Fan is ON. Current speed indicator is Low.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from Low to Medium. Medium speed indicator is active. Airflow becomes stronger than Low.	[Pending Physical Execution]	[Pending]

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC04	[Functional] Verify SPEED cycles from Medium to High.	Fan is ON. Current speed indicator is Medium.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from Medium to High. High speed indicator is active. Airflow becomes stronger than Medium.	[Pending Physical Execution]	[Pending]
TC05	[Functional / Boundary] Verify SPEED wraps from High back to Low.	Fan is ON. Current speed indicator is High.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from High to Low. Low speed indicator is active. Airflow returns to the lowest available intensity. No undefined speed state appears.	[Pending Physical Execution]	[Pending]
TC06	[Functional] Verify MODE cycles from Normal to Natural.	Fan is ON. Current mode indicator is Normal.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Normal to Natural. Natural mode indicator is active. Fan remains ON and uses the Natural wind profile.	[Pending Physical Execution]	[Pending]
TC07	[Functional] Verify MODE cycles from Natural to Sleep.	Fan is ON. Current mode indicator is Natural.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Natural to Sleep. Sleep mode indicator is active. Fan remains ON and uses the Sleep wind profile.	[Pending Physical Execution]	[Pending]

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC08	[Functional / Boundary] Verify MODE wraps from Sleep back to Normal.	Fan is ON. Current mode indicator is Sleep.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Sleep to Normal. Normal mode indicator is active. No undefined wind profile appears.	[Pending Physical Execution]	[Pending]
TC09	[Functional] Verify SWING enables horizontal oscillation.	Fan is ON. Swing is currently OFF. Fan head has safe clearance for oscillation.	1. Press SWING once. 2. Observe fan head movement and swing indicator for 10 seconds.	Horizontal oscillation starts. Swing indicator shows ON state. Fan continues producing airflow at the selected speed and mode.	[Pending Physical Execution]	[Pending]
TC10	[Functional] Verify SWING disables horizontal oscillation.	Fan is ON. Swing is currently ON and horizontal oscillation is visible.	1. Press SWING once. 2. Observe fan head movement and swing indicator for 10 seconds.	Horizontal oscillation stops. Swing indicator shows OFF state. Fan continues producing airflow at the selected speed and mode.	[Pending Physical Execution]	[Pending]
TC11	[Functional] Verify TIMER increments from no timer to the first supported timer interval.	Fan is ON. Timer is currently OFF or no timer indicator is active.	1. Press TIMER once. 2. Observe timer indicator immediately after the press.	Timer changes from OFF to the first supported auto-off interval. The corresponding timer indicator is active. Fan remains ON.	[Pending Physical Execution]	[Pending]

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC12	[Functional / Boundary] Verify TIMER cycles through all supported intervals and returns to no-timer state.	Fan is ON. Timer indicator is visible. Manufacturer-supported timer interval markings are known from the panel/manual.	1. Press TIMER repeatedly one step at a time. 2. After each press, record the active timer indicator. 3. Continue until the sequence passes the maximum supported interval.	Timer indicator advances exactly one interval per press. After the maximum supported interval, the next TIMER press returns to no-timer/OFF timer state. No skipped, duplicated, or undefined timer state appears.	[Pending Physical Execution]	[Pending]
TC13	[Functional / Boundary] Verify auto-off behavior at the shortest timer interval.	Fan is ON. Timer can be set to the shortest supported interval. Test environment allows waiting for the full interval.	1. Press TIMER until the shortest supported interval is selected. 2. Do not press any other button. 3. Wait until the selected interval elapses. 4. Observe fan state after timeout.	Fan remains ON before the interval expires. When the selected interval elapses, fan automatically changes to OFF state. Motor, airflow, and oscillation stop.	[Pending Physical Execution]	[Pending]

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC14	[Interoperability] Verify SPEED change while MODE and SWING are active.	Fan is ON. Mode is Natural or Sleep. Swing is ON. Current speed is Low or Medium.	1. Press SPEED once. 2. Observe speed indicator, mode indicator, swing indicator, airflow, and oscillation for 10 seconds.	Speed changes to the next valid speed. Existing MODE selection remains unchanged. SWING remains ON. Fan continues operating without reset or interruption.	[Pending Physical Execution]	[Pending]
TC15	[Usability] Verify control-panel button feedback and single-function response.	Fan is connected to rated AC supply. Tester can clearly view all button labels and indicators.	1. Press ON/OFF, SPEED, TIMER, SWING, and MODE one at a time under valid operating conditions. 2. After each press, observe whether only the intended function changes. 3. Confirm the associated indicator or physical behavior is recognizable.	Each button press is accepted with clear visual or physical feedback. Button labels are understandable. Each button changes only its assigned function, with no accidental activation of adjacent controls.	[Pending Physical Execution]	[Pending]

3) Verdict

- Verdict: **VALID**

- Reason: The AI successfully interpreted the product specifications for the LIFAN D-16RC-0 fan and generated 15 technically sound test cases following the requested Markdown format. While the cases were functionally correct, they lacked the specialized edge cases required to fulfill the "Analysis" level of the assignment, necessitating human intervention.

4) Reasoning (2–5 sentences)

According to ISTQB CT-AI Section 11.3, while AI is highly effective for rapid test case generation and maximizing basic coverage, it often lacks the deep domain intuition required to identify complex "corner" or edge cases without specific prompting. The output demonstrated "Functional Suitability" as defined in ISTQB CT-AI Section 4.1, accurately mapping inputs to expected results for standard operations. However, to meet the specific Bloom-AI level G9.3 (Analyse) criteria in the HW01 Requirement 3, the test suite required human augmentation to include cases the AI missed.

5) Student Fix

- Corrected artifact:

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC01	[Functional] Verify that the ON/OFF button powers on the fan from standby/off state.	Fan is assembled safely, connected to rated AC supply, and currently OFF. Control panel is visible.	1. Press the ON/OFF button once. 2. Observe fan operation and panel indicators for 5 seconds.	Fan changes to ON state. Fan motor runs and produces airflow. Control panel shows active power status. No other function requires manual activation to start basic operation.	The fan is still OFF	Failed
TC02	[Functional] Verify that the ON/OFF button powers off the fan during operation.	Fan is ON and producing airflow at any valid speed.	1. Press the ON/OFF button once. 2. Observe motor, airflow, swing state, and panel indicators for 5 seconds.	Fan changes to OFF state. Motor stops, airflow stops, oscillation stops if active, and control panel no longer indicates active operation except any allowed standby indication.	The fan is still ON	Failed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC03	[Functional] Verify SPEED cycles from Low to Medium.	Fan is ON. Current speed indicator is Low.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from Low to Medium. Medium speed indicator is active. Airflow becomes stronger than Low.	Same as expected result	Passed
TC04	[Functional] Verify SPEED cycles from Medium to High.	Fan is ON. Current speed indicator is Medium.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from Medium to High. High speed indicator is active. Airflow becomes stronger than Medium.	Same as expected result	Passed
TC05	[Functional / Boundary] Verify SPEED wraps from High back to Low.	Fan is ON. Current speed indicator is High.	1. Press SPEED once. 2. Observe the speed indicator and airflow intensity.	Speed changes from High to Low. Low speed indicator is active. Airflow returns to the lowest available intensity. No undefined speed state appears.	Same as expected result	Passed
TC06	[Functional] Verify MODE cycles from Normal to Natural.	Fan is ON. Current mode indicator is Normal.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Normal to Natural. Natural mode indicator is active. Fan remains ON and uses the Natural wind profile.	The MODE button did not function	Failed
TC07	[Functional] Verify MODE cycles from Natural to Sleep.	Fan is ON. Current mode indicator is Natural.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Natural to Sleep. Sleep mode indicator is active. Fan remains ON and uses the Sleep wind profile.	The MODE button did not function	Failed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC08	[Functional / Boundary] Verify MODE wraps from Sleep back to Normal.	Fan is ON. Current mode indicator is Sleep.	1. Press MODE once. 2. Observe the mode indicator and airflow profile for at least 10 seconds.	Mode changes from Sleep to Normal. Normal mode indicator is active. No undefined wind profile appears.	The MODE button did not function	Failed
TC09	[Functional] Verify SWING enables horizontal oscillation.	Fan is ON. Swing is currently OFF. Fan head has safe clearance for oscillation.	1. Press SWING once. 2. Observe fan head movement and swing indicator for 10 seconds.	Horizontal oscillation starts. Swing indicator shows ON state. Fan continues producing airflow at the selected speed and mode.	Same as expected result	Passed
TC10	[Functional] Verify SWING disables horizontal oscillation.	Fan is ON. Swing is currently ON and horizontal oscillation is visible.	1. Press SWING once. 2. Observe fan head movement and swing indicator for 10 seconds.	Horizontal oscillation stops. Swing indicator shows OFF state. Fan continues producing airflow at the selected speed and mode.	Same as expected result	Passed
TC11	[Functional] Verify TIMER increments from no timer to the first supported timer interval.	Fan is ON. Timer is currently OFF or no timer indicator is active.	1. Press TIMER once. 2. Observe timer indicator immediately after the press.	Timer changes from OFF to the first supported auto-off interval. The corresponding timer indicator is active. Fan remains ON.	The TIMER button did not function	Failed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC12	[Functional / Boundary] Verify TIMER cycles through all supported intervals and returns to no-timer state.	Fan is ON. Timer indicator is visible. Manufacturer-supported timer interval markings are known from the panel/manual.	1. Press TIMER repeatedly one step at a time. 2. After each press, record the active timer indicator. 3. Continue until the sequence passes the maximum supported interval.	Timer indicator advances exactly one interval per press. After the maximum supported interval, the next TIMER press returns to no-timer/OFF timer state. No skipped, duplicated, or undefined timer state appears.	The MODE button did not function	Failed
TC13	[Functional] Verify that the fan defaults to Low speed when powered ON from the standby/OFF state.	Fan is connected to rated AC supply and currently OFF. Control panel is clear and visible.	1. Press the ON/OFF button once to power ON the fan. 2. Observe the active speed indicator and initial airflow intensity.	Fan powers ON successfully. The fan speed automatically defaults to Low. The Low speed indicator is illuminated, and the initial airflow matches the Low setting.	The default speed is Medium, then change to Low	Failed

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
TC14	[Safety] Verify that the fan guard (grille) mesh layout prevents human fingers from making contact with the rotating blades.	Fan is fully assembled, powered ON, and the blades are rotating at High speed. A standard test finger or safe visual inspection tool is available.	1. Gently attempt to pass a safe finger-sized probe through the front and rear fan guards at various gap locations. 2. Verify if the probe can reach the rotating blades.	The fan guard gaps are narrow enough to block the probe entirely. It is physically impossible for a finger to touch or come into dangerous proximity with the rotating fan blades from any angle.	Same as expected result	Passed
TC15	[Appearance / UI] Verify the legibility, print quality, and clarity of all text and markings on the fan body.	Fan is placed in a well-lit test environment. Tester inspects the device from a standard user distance.	1. Read and inspect all text on the fan body (Brand logo "LIFAN", Model "D-16RC-0", and button labels: SPEED, TIMER, SWING, MODE, ON/OFF). 2. Check for printing defects, fading, or misspellings.	All text, symbols, and labels are perfectly legible, sharply printed, and easy to read. Font alignment is consistent, and there are no signs of smudging, faded ink, or spelling errors.	Same as expected result	Passed

- What changed: Remove 3 last test cases with edge cases.

AI Accuracy Summary (required at end)

- Total artifact entries: 6
- VALID: 3 (50%)
- INVALID: 3 (50%)
- INCOMPLETE: 0 (0%)

Conclusion

- When AI should be used for this HW:
 - Research for information but need validation.
 - Format the documents.
- When AI should not be used for this HW:
 - Do screen shot

10) Appendix B — Prompt Log

1. Requirement 1

- Timestamp: 16:12 01/06/2026
- Model: GPT-5.5
- Prompt: "Read the requirements\2026.HW01.Jobs.Defects.PhysicalProduct_En.pdf, complete Requirement 1 - QA/QC Job Market 2026+ using chrome-devtools-mcp. Focus on LinkedIn first. After that, complete the Requirement 1 section in report.md, using templates\hw01-job-postings-template.md."

```
> Read the requirements\2026.HW01.Jobs.Defects.PhysicalProduct_En.pdf, complete Requirement 1 - QA/QC Job Market 2026+ using chrome-devtools-mcp. Focus on LinkedIn first. After that, complete the Requirement 1 section in report.md, using templates\hw01-job-postings-template.md.
```

2. Copy exact job description and reformat Requirement 1 section.

- Timestamp: 22:58 02/06/2025
- Model: GPT-5.4
- Prompt: "In the Requirement 1 section of report.md, using chrome-devtools-mcp@latest, visit all the websites of the jobs to copy the full job description and put them into quotation (>). Reformat other jobs section to be similar as the posting 01."

```
> In the Requirement 1 section of report.md, using chrome-devtools-mcp@latest, visit all the websites of the jobs to copy the full job description and put them into quotation (>). Reformat other jobs section to be similar as the posting 01.
```

3. Requirement 1: Rewrite skills + ai impact

- Timestamp: 15:35 03/06/2026
- Model: GPT-5.5
- Prompt: "In requirement 01 of report.md, for each job, based on the job description, rewrite the Required skills and AI Impact Analysis."

```
> In requirement 01 of report.md, for each job, based on the job description, rewrite the Required skills and AI Impact Analysis.
gpt-5.5 20% ~\Documents\software-testing\HW01 validation (11) +101 -77 In report.md
>> accept edits on (shift+tab to cycle)
```

4. 20 defects

- Timestamp: 16:18 03/06/2026

- Model: Gemini 3.1 Pro
- Prompt: "Find 20 software defects publicized from cve.org between 2022 and 2026:
 - Mandatory: ≥ 5 defects related to AI/LLM (hallucination, prompt injection, bias).
 - Each defect includes: source link, description, severity, consequences, solution."

Find 20 software defects publicized from cve.org between 2022 and 2026:

- Mandatory: ≥ 5 defects related to AI/LLM (hallucination, prompt injection, bias).

- Each defect includes: source link, description, severity, consequences, solution.



5. QA/QC role mindmap

- Timestamp: 08:15 04/06/2026
- Model: Gemini 3.1 Pro
- Prompt: "Using the PDF file I provided, generate the QA/QC role mindmap in mermaid code."

PDF

ISTQB_CT-
AI_Syllabus_v
1.0

Using the PDF file I provided, generate the QA/QC role mindmap in mermaid code.



6. Generate 15 testcases

- Timestamp: 19:26 04/06/202
- Model: GPT 5.5
- Prompt: Below

Act as an expert QA/QC Test Engineer. I need to design test cases for a physical product for Requirement 3 section in report.md.

Product Specification:

- Product Type: Domestic Stand Fan (Model: LIFAN D-16RC-0)
- Control Interface: It has 5 electronic/soft buttons on the control panel:
 1. ON/OFF (1 single button to toggle power status)
 2. SPEED (Cycles through wind speeds: Low - Medium - High)
 3. TIMER (Increments time delay to turn off the fan automatically)
 4. SWING (Toggles horizontal oscillation on and off)
 5. MODE (Cycles through wind profiles: Normal, Natural, Sleep)

Task Requirements:

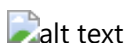
- Generate exactly 15 comprehensive test cases covering Functional, Usability, and Interoperability scenarios based on the interface above.
- Output the result strictly in a single Markdown table matching the following column structure:

TC ID	Objective	Input / Pre-conditions	Execution Steps	Expected Result	Actual Result	Verdict
-------	-----------	------------------------	-----------------	-----------------	---------------	---------

Instructions for columns:

1. TC ID: Use numbering format starting from TC01 to TC15.
2. Actual Result: Leave this column completely empty (or put "[Pending Physical Execution]") so I can fill it out manually after testing the real device.
3. Verdict: Leave this column empty (or put "[Pending]").
4. Ensure all descriptions, steps, and expected results are written in technical English and highly precise for physical hardware testing.

Write them into report.md in the Requirement 3 section.



11) Appendix C — AI Disclosure Form

AI-03 — AI Disclosure Form Template

Student Information

- Student ID: 23127075
- Student Name: Le Trung Kien
- Course / Exercise: Software Testing / HW01
- Date: 04/06/2026

Disclosure Statement

I disclose that AI tools were used in this assignment under course policy. I reviewed, validated, and corrected AI outputs before submission.

AI Tools Used

Tool	Purpose	Sections Affected
GPT	Research and format	All
Gemini	Research	All

Mandatory Declaration (edit placeholders)

"All the test cases was initially generated by GPT-5.5; I reviewed and modified last 3 testcases, added edge cases 13 14 16; deflects evaluation was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category below."

Prohibited-Artifacts Compliance

- ☒ Device photo with student ID in same frame was produced by me (not AI generated)
- ☒ Execution videos with my own voice narration were produced by me
- ☒ Job-posting screenshots with my account/display name were captured by me
- ☒ Prompt log with timestamps is complete and authentic

Signature

- Student signature: **tkin**
- Date: **04/06/2026**

12) Appendix D — AI Privacy & Responsible Use Checklist

AI-05 — AI Privacy & Responsible Use Checklist Template

Student Information

- Student ID: **<...>**
- Student Name: **<...>**
- Exercise: **HW01**
- Date: **<dd/mm/yyyy>**

Privacy & Responsible Use Checklist

Data Handling

- ☒ I did not upload sensitive personal data beyond what is required for the assignment
- ☒ I masked device serial number middle 4 characters
- ☒ I reviewed AI outputs for fabricated or unsafe instructions

Academic Integrity

- ☒ I did not submit AI output without verification
- ☒ I included full prompt log with timestamps
- ☒ I included AI Audit Report entries for AI-generated artifacts
- ☒ I clearly marked which sections were AI-assisted vs written by me

Anti-Cheat Compliance

- ☒ Device photo includes my student ID in same frame
- ☒ Test execution videos contain my own voice narration
- ☒ Job posting screenshots show my account/display name
- ☒ Evidence artifacts are authentic and reproducible

Quality Controls

- ☒ I validated AI-generated test cases against ISTQB/course materials
- ☒ I identified and documented AI mistakes/bias/hallucination
- ☒ I added student-generated edge cases AI missed

Acknowledgement

I confirm this checklist is accurate and complete.

- Signature: **tkin**
- Date: **04/06/2026**